Space Exploration & Aerospace Engineering

Part VII

Space Environmental Science

The Sun drives a violent 'weather' system that reaches every satellite and power grid on Earth. Part VII explains solar activity, geomagnetic storms, the magnetosphere, radiation belts and the ionosphere — and the missions and instruments that study them.

Chapters 22–23 · pages 396–429 of the complete manual
First Edition · Current to October 2026 · Written in plain language with deep technical detail

ENGINEERING REFERENCE MANUAL

Space Exploration & Aerospace Engineering

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First Edition · Information current to October 2026
A4 format · Fourteen Parts · Fifty-three chapters · Ten appendices
Version 1.0 · October 2026

About this edition

Space Exploration & Aerospace Engineering — An Engineering Reference Manual. First Edition, Version 1.0, October 2026.

This manual was written to be read by students, engineers, analysts, managers and curious readers. Each topic starts with plain-language explanations and builds up to the equations, numbers and design trade-offs that practising engineers use. Sentences are kept short and every technical term is explained in plain words when it first appears; the glossary (Appendix H) and the list of acronyms (Appendix G) collect them. Facts, mission status and programme milestones are current to October 2026; space programmes change quickly, so dates of future events should be read as plans, not certainties.

Figures, photographs and data

Diagrams, charts and graphs were drawn specifically for this manual from published physical relationships and public data. Photographs are from the NASA Image and Video Library and are used under NASA's media usage guidelines (NASA material is generally not subject to copyright in the United States); individual credits appear with each image and in Appendix J. NASA does not endorse this publication. Numerical values (masses, power levels, performance) are representative published figures and may differ slightly between sources and mission phases.

Safety and responsibility

This is an educational reference. It is not a design standard, a flight rule, or a substitute for the requirements, analyses and verification processes of a qualified engineering organisation. Chapters dealing with military and intelligence systems use only openly published information and describe principles, not operational or classified details.

How numbers are written

SI units are used throughout, with common aerospace units (nautical miles, feet, pounds-force) shown where the industry uses them. Large numbers use thin spacing (e.g., 35 786 km). "≈" means approximately, "~" means order-of-magnitude. Dates follow the ISO format (YYYY-MM-DD) in tables.

Typography and production

Set in Source Serif 4, Inter and JetBrains Mono. Equations typeset with MathJax. Produced from Markdown sources with an automated pipeline (pandoc, Chromium, pypdf); the sources and build scripts accompany the PDF.

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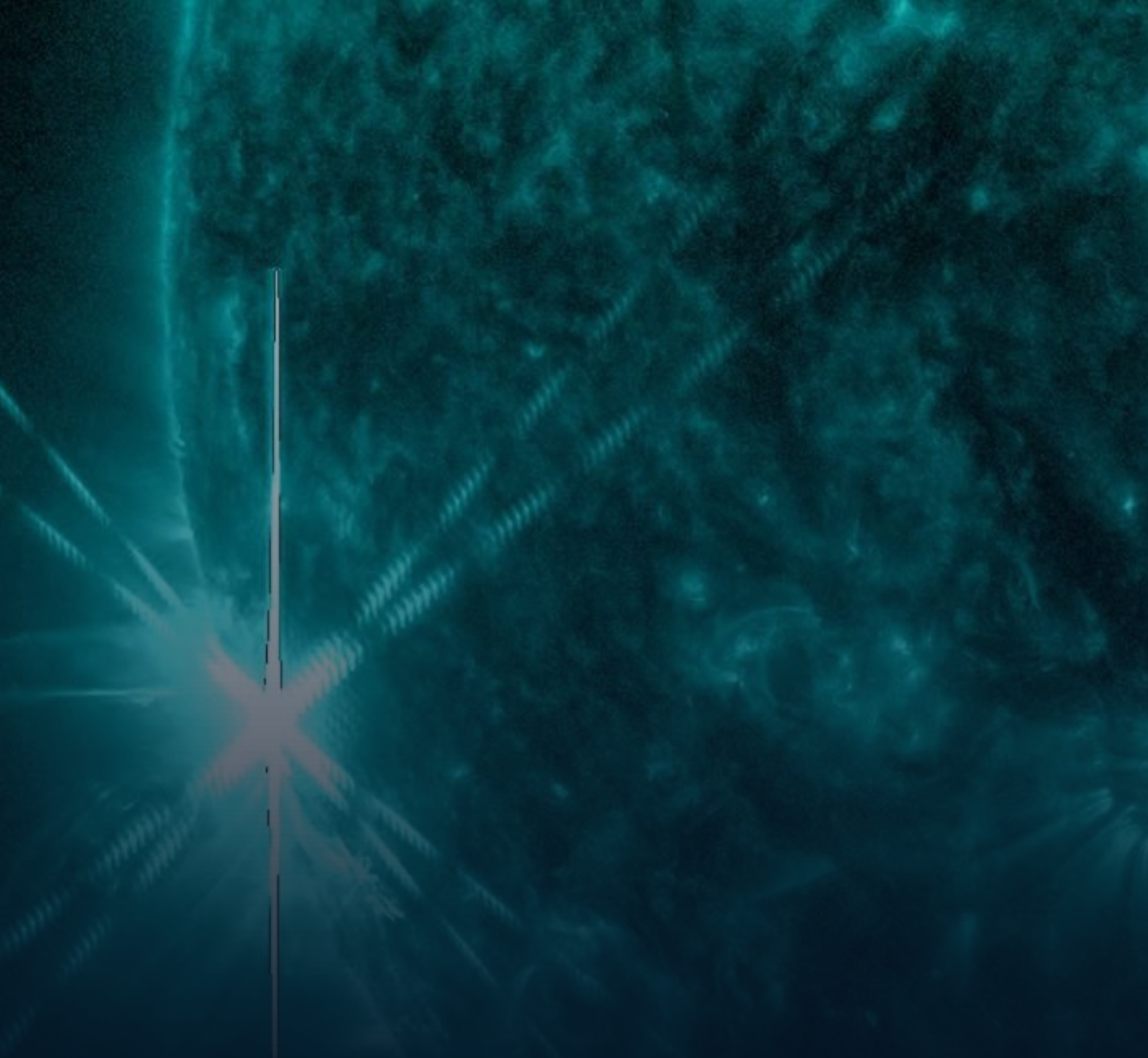
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This volume is one Part of *Space Exploration & Aerospace Engineering — An Engineering Reference Manual*. Page numbers, chapter numbers and figure numbers are the same as in the complete manual, so cross-references to other chapters (for example “Chapter 24”) and to the appendices refer to the complete book.



PART VII

Space Environmental Science

The Sun drives a violent 'weather' system that reaches every satellite and power grid on Earth. Part VII explains solar activity, geomagnetic storms, the magnetosphere, radiation belts and the ionosphere — and the missions and instruments that study them.

22 Space Weather

23 Magnetospheric and Ionospheric Physics

Image: NASA/GSFC

22 CHAPTER Space Weather

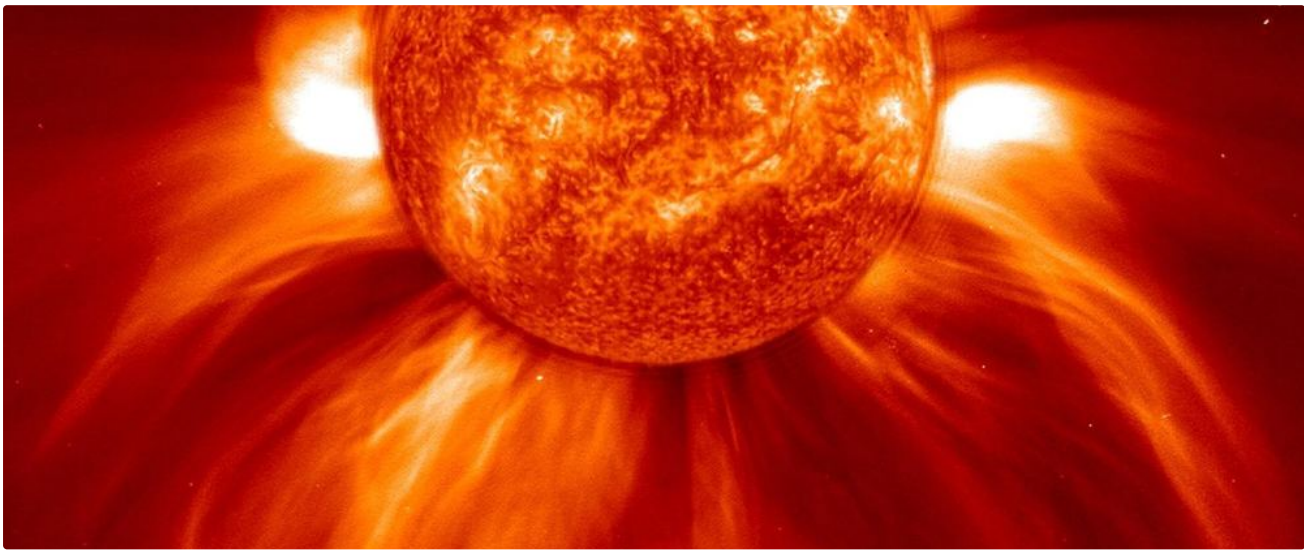


Image: NASA/GSFC

The Sun is a variable star whose eruptions — flares, coronal mass ejections and storms of energetic particles — can disrupt satellites, navigation, radio communications, aviation and power grids on Earth and endanger astronauts. This chapter explains the physics of space weather from the solar dynamo and the 11-year sunspot cycle to the arrival of a storm at Earth, using real records of the sunspot number since 1749 and of geomagnetic activity since 1932. It describes the great historical storms, the May 2024 "Gannon" storm — the strongest in two decades — and the effects of space weather on technology, with engineering mitigations for each. It then covers the observatories, from SOHO to the new L1 monitors launched in 2025, and the models and services that forecast space weather today.

IN THIS CHAPTER

22.1 What Is Space Weather?
22.2 The Sun: A Variable Star
22.3 Solar Eruptions and the Solar Wind
22.4 Space Weather at the Earth
22.5 Effects on Technology and People
22.6 Observing and Forecasting Space Weather
22.7 Designing for Space Weather

22.1 What Is Space Weather?

Space weather describes the conditions in space — on the Sun, in the solar wind, and in the Earth's magnetosphere, ionosphere and thermosphere — that can affect technology and human life. Just as terrestrial weather has its storms, space weather has its eruptions. There are **solar flares** that flood the sunlit side of the Earth with X-rays in minutes, and **solar energetic particle**

(SEP) events that fill near-Earth space with high-energy protons. There are also **coronal mass ejections (CMEs)** — billion-tonne clouds of magnetised plasma that can trigger **geomagnetic storms** when they reach the Earth one to four days later.

Modern society is increasingly exposed. Satellites can be damaged or lost, and GNSS positioning degrades. High-frequency radio used by aviation and maritime users blacks out, and airline passengers and crew receive extra radiation on polar routes. Long power lines and pipelines carry damaging induced currents. Several governments, including those of the UK and the USA, list severe space weather among their national risks. Insurance studies have estimated that a storm comparable to the largest on record could cause economic losses of hundreds of billions to trillions of dollars. These losses would come mainly through prolonged power outages.

22.2 The Sun: A Variable Star

22.2.1 Structure

The Sun is a sphere of hot plasma 1.39 million km across, powered by nuclear fusion in its core at about 15 million K (Chapter 2). Energy travels outward by radiation through the **radiative zone** and then by convection through the **convection zone**, the outer 30 % of the radius. The visible surface, the **photosphere**, has a temperature of about 5 800 K. Above it lie the thin **chromosphere** and the tenuous **corona**, which is mysteriously heated to 1–2 million K and continuously expands into space as the **solar wind**. NASA's **Parker Solar Probe**, which on 24 December 2024 passed just 6.1 million km from the Sun's surface at a record 192 km/s, is investigating how the corona is heated and the wind accelerated.

22.2.2 Magnetic Activity and the Solar Cycle

The Sun's activity is driven by its **magnetic field**, generated by a dynamo. In this dynamo, differential rotation (the equator rotates in about 25 days, the poles in about 35) and convection stretch, twist and amplify the field. Where bundles of strong magnetic field emerge through the photosphere, they inhibit convection and create cooler, darker **sunspots** (Figure 22.3), surrounded by active regions where flares and CMEs originate.

The number of sunspots rises and falls in a cycle of about **11 years** (Figure 22.1), during which the Sun's global magnetic field reverses; the full magnetic cycle therefore lasts about 22 years. Systematic sunspot counts began in the 18th century; the cycle that began in 1755 is numbered Cycle 1. Cycles vary greatly in strength: Cycle 19, peaking in 1958, was the strongest recorded, while from about 1645 to 1715 — the **Maunder Minimum** — sunspots almost disappeared.

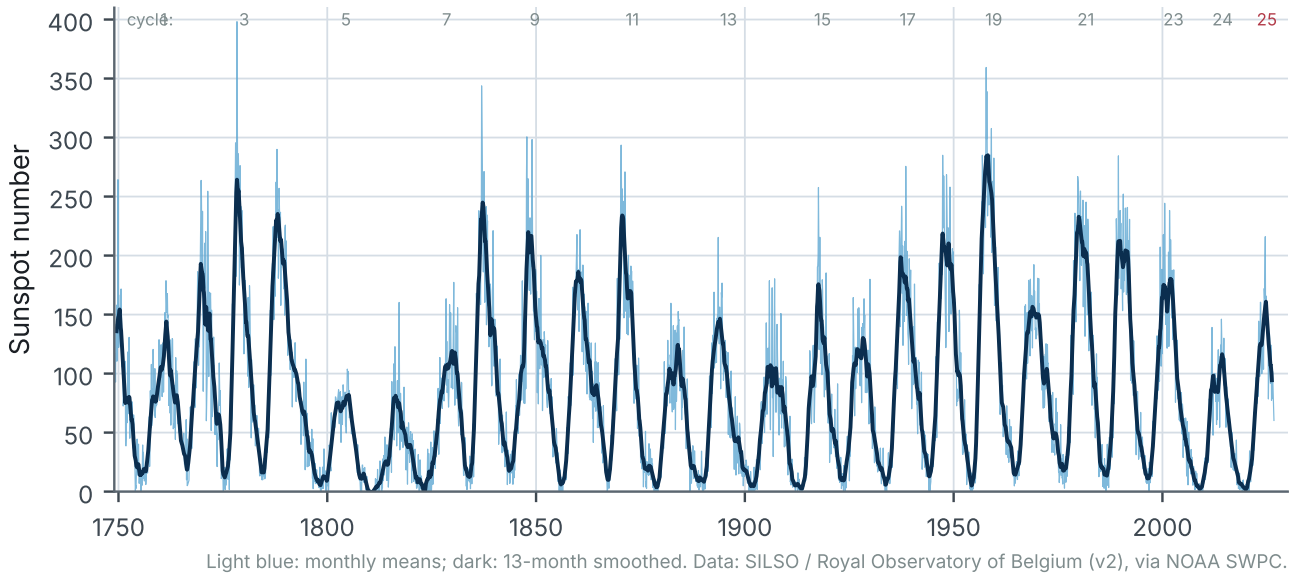


FIGURE 22.1 The sunspot number since 1749. Cycle numbers are shown at the top. The strength and length of cycles vary considerably.

Solar Cycle 25 began in December 2019. A NOAA–NASA expert panel predicted a weak cycle peaking at a smoothed sunspot number of about 115, similar to Cycle 24. Instead, activity rose faster and higher, reaching a smoothed maximum of about 161 in October 2024 (Figure 22.2). The declining phase, now under way, is historically when many of the largest geomagnetic storms occur, driven both by CMEs and by high-speed solar-wind streams from coronal holes.

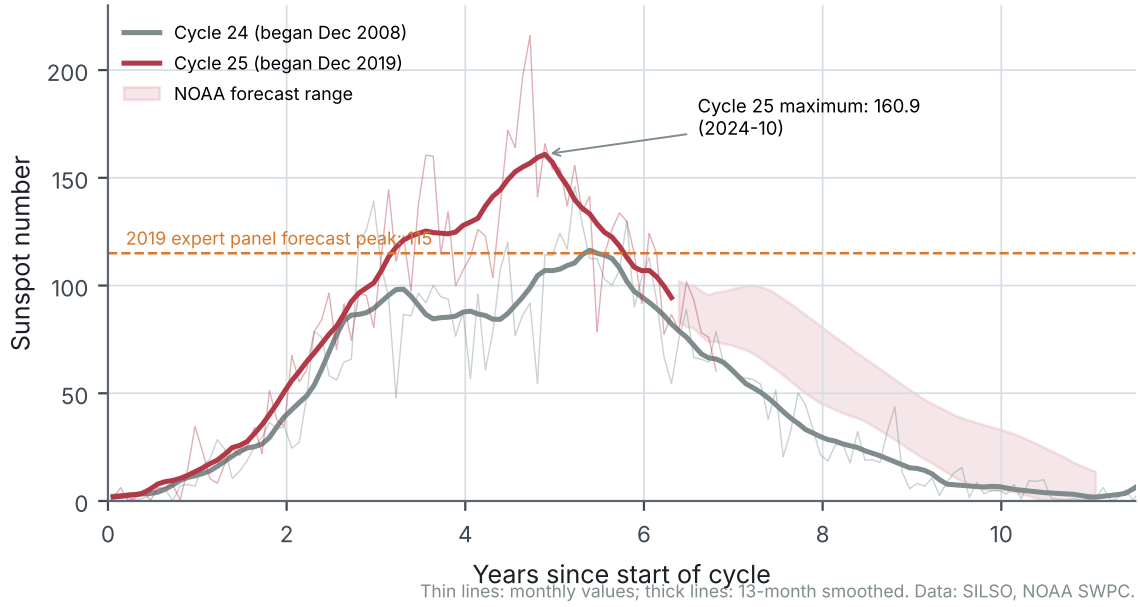


FIGURE 22.2 Solar Cycle 25 compared with Cycle 24 and with the 2019 panel forecast.

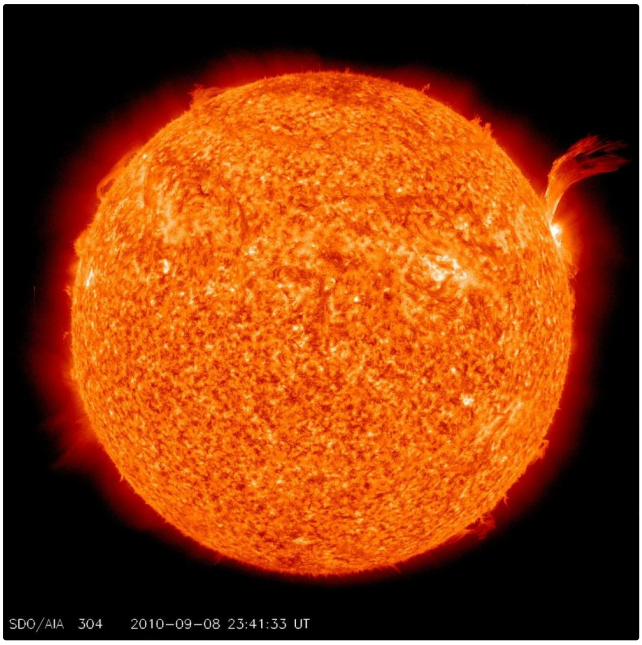


FIGURE 22.3 The Sun seen by NASA's Solar Dynamics Observatory in extreme ultraviolet light on 8 September 2010, as a C3-class flare erupted from an active region near the edge of the disk. Active regions sit above sunspot groups, where strong magnetic fields break through the surface. Image: NASA/GSFC.

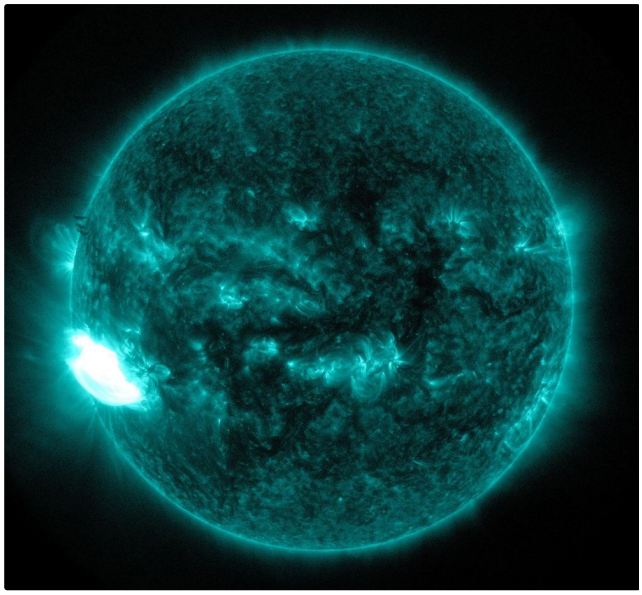


FIGURE 22.4 An X-class solar flare (bright region at left) on 19 October 2014, observed by SDO in extreme ultraviolet light. Image: NASA/GSFC.

22.3 Solar Eruptions and the Solar Wind

22.3.1 Solar Flares

A **solar flare** is a sudden release of magnetic energy in an active region — up to about 10^{25} J in minutes to hours. It heats plasma to tens of millions of kelvin and accelerates particles. Flares emit across the spectrum from radio waves to gamma rays; their X-ray and extreme-ultraviolet radiation reaches the Earth in about 8 minutes and immediately increases ionisation in the dayside ionosphere. Flares are classified by their peak soft X-ray flux at 0.1–0.8 nm, measured by GOES satellites. The classes are A, B, C, M and X, each ten times stronger than the last. An X1 flare corresponds to 10^{-4} W/m^2. The largest flare of the space age, on 4 November 2003, saturated the detectors and has been estimated at about X40.

22.3.2 Coronal Mass Ejections

A **coronal mass ejection** expels 10^{11}–10^{13} kg of magnetised plasma from the corona at speeds from about 300 to more than 3 000 km/s (Figure 22.5). CMEs are observed with **coronagraphs**, which block the bright solar disc to reveal the faint corona. When a fast CME is directed towards the Earth, it typically arrives in one to three days; the fastest have crossed the 150 million km in under 18 hours. The geomagnetic impact of a CME depends above all on the direction of its internal magnetic field. A strong **southward** field component (B_z negative) connects efficiently with the Earth's northward field at the dayside magnetopause. This opens the magnetosphere to the solar wind's energy (Chapter 23).

FIGURE 22.5 A coronal mass ejection erupting from the Sun's limb on 17–18 June 2015, imaged by SDO in extreme ultraviolet light. Image: NASA/GSFC.

22.3.3 Solar Energetic Particles

Flares and the shocks driven by fast CMEs accelerate protons, electrons and heavier ions to energies of millions to billions of electronvolts. These **solar energetic particles** travel along the interplanetary magnetic field and can reach the Earth within tens of minutes. They raise the flux of energetic protons near the Earth by many orders of magnitude for hours to days. The most energetic events produce **ground-level enhancements** detected by neutron monitors at the Earth's surface. SEPs are the main acute radiation hazard for astronauts beyond low Earth orbit and a major cause of single-event effects in spacecraft electronics.

22.3.4 The Solar Wind

Between eruptions, the corona continuously emits the **solar wind**: about 5 protons and electrons per cubic centimetre at 1 AU. It flows at 300–500 km/s ("slow wind") or 600–800 km/s ("fast wind" from **coronal holes**), carrying the interplanetary magnetic field of about 5 nT. Where fast wind catches up with slow wind, **corotating interaction regions** form; because coronal holes can persist for months, they produce geomagnetic disturbances that recur every 27 days as the Sun rotates.

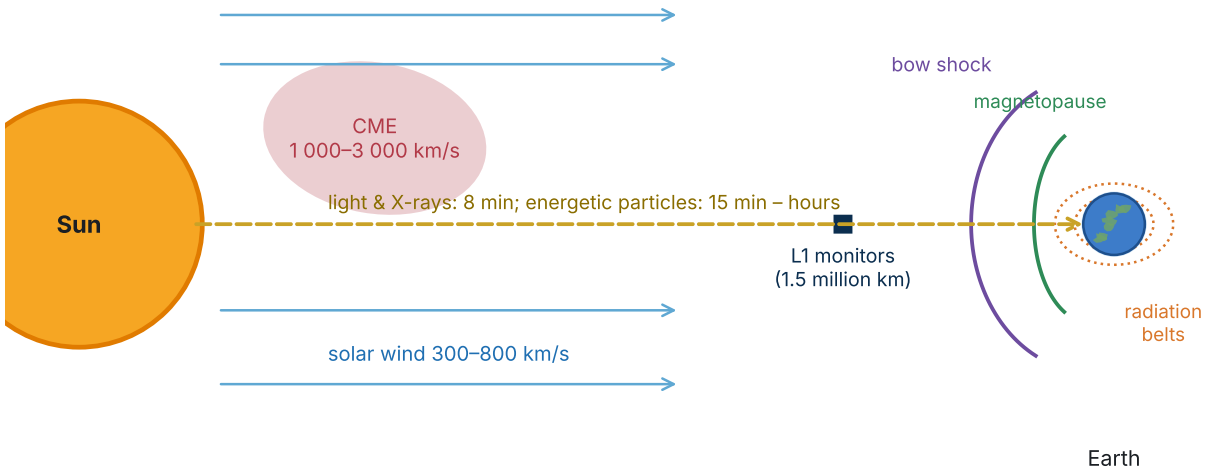


FIGURE 22.6 The chain of space weather from the Sun to the Earth (schematic, not to scale).

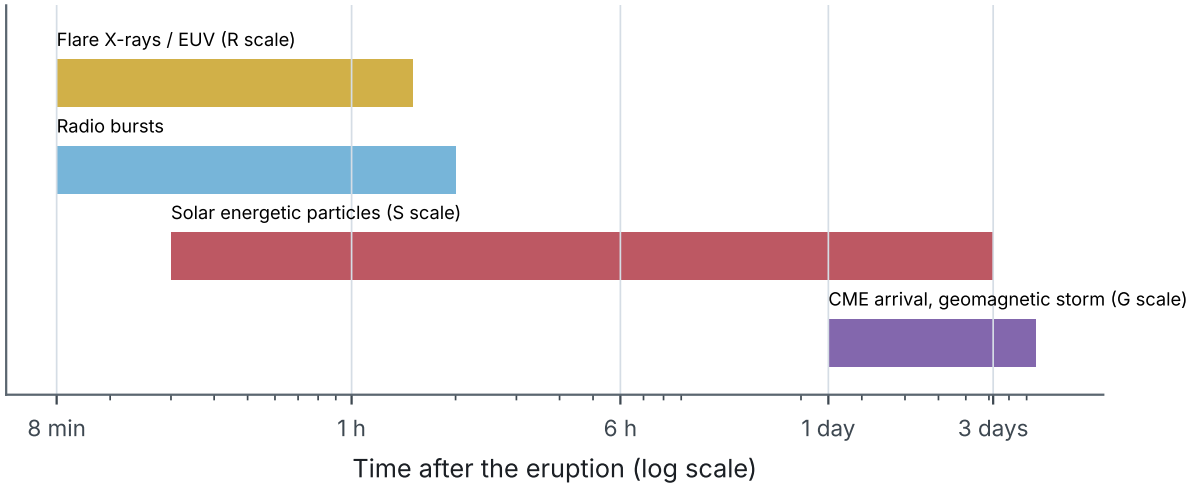


FIGURE 22.7 Typical timing of the effects of a major solar eruption at the Earth.

22.4 Space Weather at the Earth

22.4.1 Geomagnetic Storms and Indices

When a CME or fast stream with a southward magnetic field reaches the Earth, energy pours into the magnetosphere, driving a **geomagnetic storm**. The ring current around the Earth intensifies and the auroral ovals expand towards the equator. Strong currents flow in the polar ionosphere, and the thermosphere heats and expands. Storms are measured by **geomagnetic indices** derived from networks of ground magnetometers:

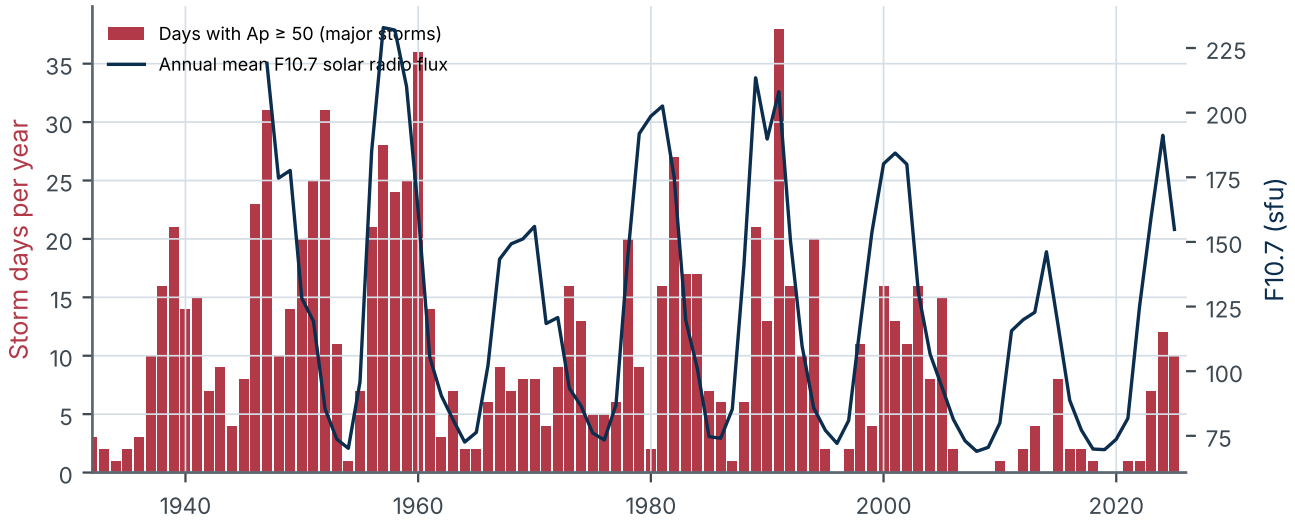
- **Kp** (0 to 9, quasi-logarithmic) measures global disturbance in three-hour intervals; **Ap** is its daily linear equivalent.
- **Dst** (in nanotesla) measures the depression of the horizontal magnetic field at low latitudes due to the ring current; major storms reach −200 to −600 nT.
- **AE** measures the intensity of the auroral electrojets.

NOAA's **space weather scales** (Table 22.1) translate these measurements into five levels of severity for geomagnetic storms (G), solar radiation storms (S) and radio blackouts (R).

TABLE 22.1 NOAA space weather scales (pfu = particle flux units, particles cm^{-2} s^{-1} sr^{-1}).

Level	Geomagnetic storm (G)	Solar radiation storm (S)	Radio blackout (R)	Approximate frequency per 11-year cycle
1 Minor	Kp 5	≥ 10 pfu (≥ 10 MeV protons)	M1 flare	G1: ≈ 1 700; S1: ≈ 50; R1: ≈ 2 000
2 Moderate	Kp 6	≥ 10^2 pfu	M5	G2: ≈ 600; S2: ≈ 25; R2: ≈ 350
3 Strong	Kp 7	≥ 10^3 pfu	X1	G3: ≈ 200; S3: ≈ 10; R3: ≈ 175
4 Severe	Kp 8 (including 9−)	≥ 10^4 pfu	X10	G4: ≈ 100; S4: ≈ 3; R4: ≈ 8
5 Extreme	Kp 9	≥ 10^5 pfu	X20	G5: ≈ 4; S5: < 1; R5: < 1

The long record of geomagnetic activity (Figure 22.8) shows that major storms are concentrated around solar maximum and in the declining phase of each cycle, but that severe storms can occur at almost any time.



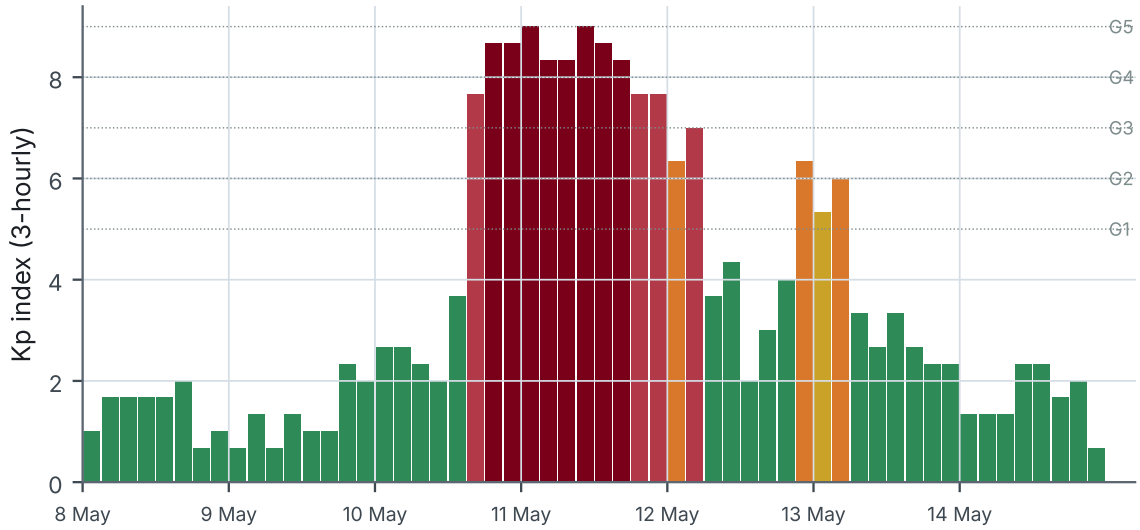
Data: GFZ Potsdam Kp/Ap index and F10.7 flux (Penticton/Ottawa). Storms are most frequent in the declining phase of each cycle.

FIGURE 22.8 Number of days per year with major geomagnetic storms (Ap ≥ 50) and the annual mean solar radio flux, 1932–2025.

22.4.2 The May 2024 "Gannon" Storm

From 8 May 2024, a cluster of sunspots (region 13664) produced a series of X-class flares and at least six Earth-directed CMEs. They arrived on 10 May and combined to produce the first **G5 (extreme)** geomagnetic storm since 2003 (Figure 22.9), with Kp reaching 9 and Dst falling to about −410 nT. Aurorae were seen as far south as Florida, northern India and Namibia, and photographed by millions of people with smartphones. The storm, named after the solar physicist Jennifer Gannon, caused GNSS-guided tractors to lose accuracy during the planting season in North America. It also degraded HF communications and increased drag on thousands

of satellites, causing mass manoeuvres and temporary loss of tracking. In power grids it induced currents, which operators managed without major failures. Further severe storms followed in October 2024, November 2025 and January 2026.



Data: GFZ Potsdam definitive Kp. The 10–11 May 2024 'Gannon storm' reached Kp 9 (G5), the strongest since 2003.

FIGURE 22.9 Three-hourly Kp index during the May 2024 geomagnetic storm (8–14 May).

22.4.3 Great Storms of the Past

TABLE 22.2 Notable space weather events.

Date	Event	Notable effects
1–2 Sept 1859	Carrington event	First observed white-light flare; telegraph systems failed and caught fire; aurora seen near the equator; Dst estimated −850 to −1 050 nT
13–15 May 1921	“New York Railroad” storm	Fires in telegraph and railway signalling stations; possibly comparable to Carrington
4 Aug 1972	Solar particle event and storm	Between Apollo 16 and 17 — a lethal dose risk for astronauts on the Moon; detonated sea mines off Vietnam
13 Mar 1989	Quebec storm	Hydro-Québec grid collapsed in 92 s, leaving 6 million people without power for 9 hours; tracking of thousands of satellites lost
29–31 Oct 2003	Halloween storms	X17 and X10 flares (and an estimated X40 on 4 Nov); satellite anomalies and a lost Japanese satellite (ADEOS-II); power outage in Malmö, Sweden; aircraft rerouted; WAAS vertical guidance unavailable for ≈ 30 hours
23 Jul 2012	Near miss	Extreme CME observed by STEREO-A missed the Earth by about nine days of solar rotation
3–4 Feb 2022	Minor storm, major loss	Increased drag prevented 38 of 49 newly launched Starlink satellites from raising their orbits

Date	Event	Notable effects
10–11 May 2024	Gannon storm	First G5 since 2003; worldwide aurorae; GNSS and satellite drag effects

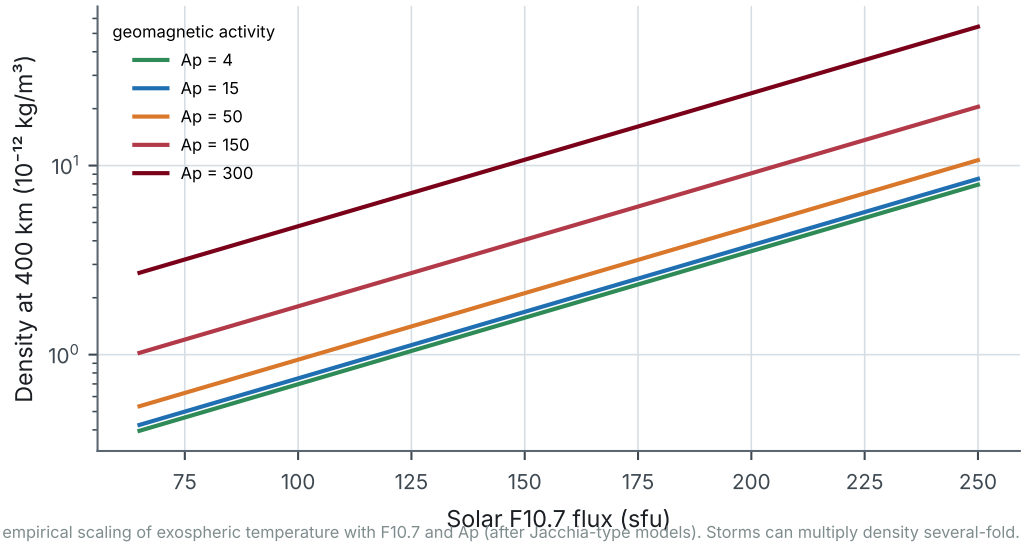
Even larger events have occurred. Tree rings and ice cores record sudden increases in radiocarbon and beryllium-10 in 774–775 and 993–994 CE (**Miyake events**). These are attributed to solar particle storms perhaps ten times stronger than any observed in the space age.

22.5 Effects on Technology and People

22.5.1 Satellites

Space weather affects spacecraft in several ways (Chapter 2):

- **Surface charging**: during storms, hot (keV) electrons injected into the inner magnetosphere charge spacecraft surfaces to thousands of volts, especially in the night-side GEO region. Discharges ("arcs") can damage solar arrays and electronics or cause phantom commands.
- **Internal (deep dielectric) charging**: high-energy (MeV) electrons of the outer radiation belt, which can increase dramatically for days after a storm, penetrate shielding and accumulate in cable insulation and circuit boards until they discharge. Internal charging is blamed for the failures of Canada's Anik E1 and E2 communications satellites in January 1994. It is also blamed for the failure of Galaxy 15 in 2010, which drifted out of control for eight months as a "zombie satellite" still relaying signals.
- **Single-event effects**: solar energetic particles and galactic cosmic rays cause bit flips, latch-ups and burnouts in electronics; star trackers become blinded by particle hits during SEP events.
- **Total ionising dose and displacement damage**: SEP events can deliver in days the dose expected over years, degrading solar cells and sensors.
- **Atmospheric drag**: heating of the thermosphere by storms increases density at satellite altitudes several-fold (Figure 22.10). In February 2022, a minor storm prevented 38 of 49 Starlink satellites, released at a low parking altitude, from raising their orbits; they re-entered within days. During major storms, thousands of objects change their orbits unpredictably, degrading the space catalogue and conjunction assessment (Chapter 21).



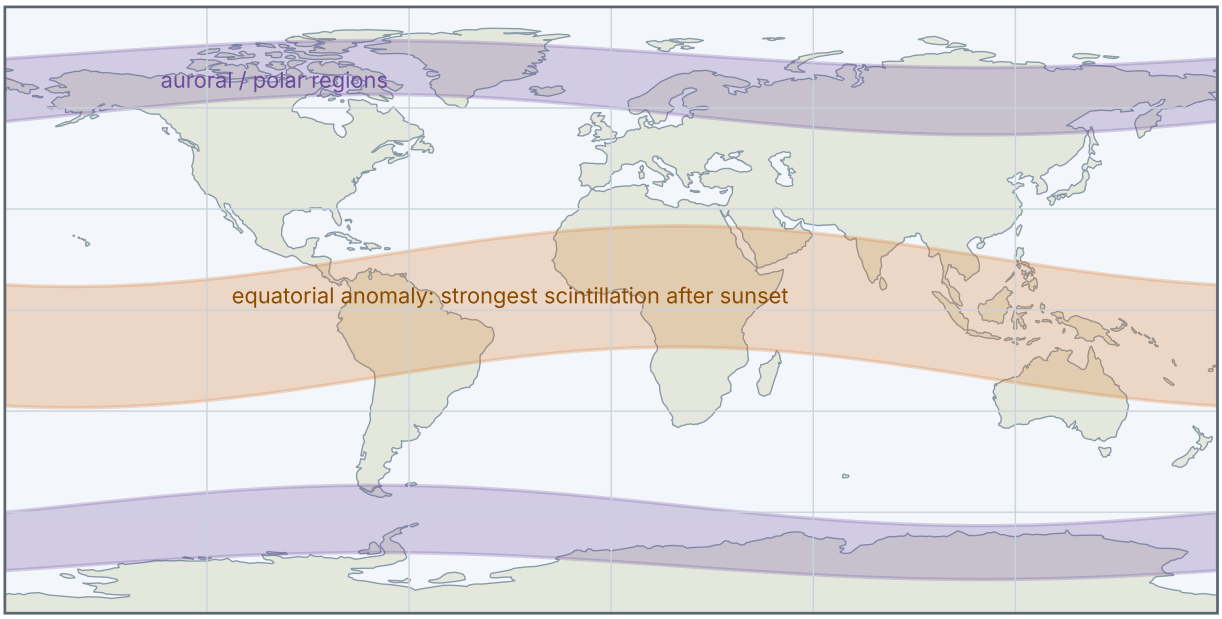
Illustrative empirical scaling of exospheric temperature with F10.7 and Ap (after Jacchia-type models). Storms can multiply density several-fold.

FIGURE 22.10 Illustrative density of the thermosphere at 400 km as a function of solar EUV activity (F10.7) and geomagnetic activity (Ap). Both solar maximum and storms greatly increase satellite drag.

22.5.2 GNSS, Radio and Aviation

Disturbances of the ionosphere affect radio signals passing through it (Chapter 23):

- **GNSS errors**: increases and steep gradients of total electron content during storms produce positioning errors of tens of metres for single-frequency users. They also degrade augmentation systems. During the Halloween 2003 storms, WAAS could not provide vertical guidance for about 30 hours.
- **Scintillation**: small-scale irregularities in the ionosphere cause rapid fluctuations of signal amplitude and phase, which can make receivers lose lock. This is common after sunset near the geomagnetic equator and in the auroral and polar regions (Figure 22.11).
- **HF radio blackouts**: flare X-rays increase ionisation in the D region, absorbing HF signals on the dayside for minutes to hours. Energetic protons cause **polar cap absorption** lasting days, forcing airlines to reroute polar flights that rely on HF communications.
- **Radiation exposure**: during SEP events, aircrew and passengers on high-latitude, high-altitude flights receive increased radiation doses; aviation authorities issue advisories, and some flights are lowered or rerouted.



Schematic regions of frequent ionospheric scintillation affecting GNSS and satellite communications (after NOAA SWPC).

FIGURE 22.11 Regions where ionospheric scintillation most often affects GNSS and satellite communications (schematic).

22.5.3 Power Grids and Other Ground Systems

Rapid changes in the Earth's magnetic field during storms (large dB/dt) induce **geoelectric fields** in the conducting ground — typically 1–10 V/km in severe storms, more over resistive rock (Figure 22.12). These fields drive **geomagnetically induced currents (GICs)** through long conductors grounded at both ends: high-voltage power lines, pipelines, railways and submarine cables. In power grids, GICs of tens to hundreds of amperes flow through transformer windings into the ground, saturating transformer cores, generating harmonics, causing overheating and tripping protective relays. The March 1989 storm collapsed the Hydro-Québec grid in about 90 seconds and damaged a large transformer at the Salem nuclear plant in New Jersey.

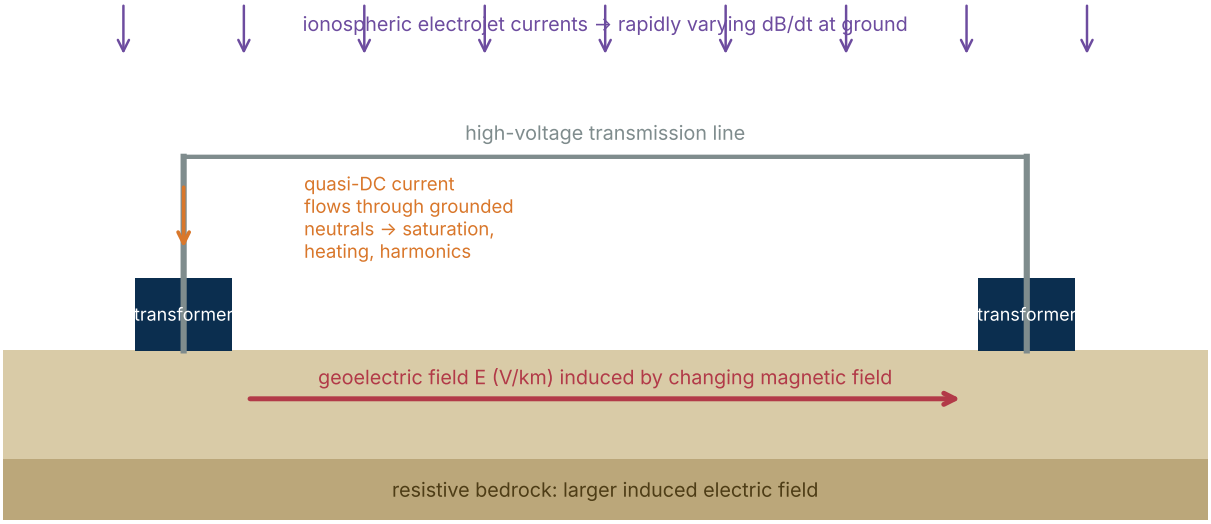


FIGURE 22.12 Geomagnetically induced currents. Ionospheric currents cause rapid magnetic variations that induce an electric field in the ground and drive quasi-DC currents through grounded transformers.

Grid operators now receive storm warnings and run models of GIC flows. They can reduce loading on vulnerable transformers or reconfigure the network. Regulators in North America and Europe require utilities to assess their vulnerability to benchmark extreme storms.

22.5.4 Astronauts

Outside the protection of the Earth's magnetosphere, astronauts are exposed to SEP events that could deliver dangerous doses in hours. Had the August 1972 event occurred during an Apollo moonwalk, the astronauts could have suffered acute radiation sickness. Crewed spacecraft therefore provide **storm shelters** — areas surrounded by water, food, equipment and propellant — and missions rely on real-time warnings (Chapter 30). Artemis II's Orion spacecraft, flown around the Moon in April 2026, carried radiation sensors and a shelter procedure.

22.6 Observing and Forecasting Space Weather

22.6.1 Observatories

Space weather is monitored by a fleet of solar and heliospheric spacecraft and by ground networks (Table 22.3):

TABLE 22.3 Space weather observatories (selected).

Mission	Agencies	Launch	Location	Role
SOHO	ESA / NASA	1995	L1	LASCO coronagraph: the main CME monitor for 30 years
ACE	NASA	1997	L1	Solar-wind particles and magnetic field
STEREO-A	NASA	2006	Heliocentric orbit	Views of the Sun and CMEs from the side
SDO	NASA	2010	Geosynchronous	High-resolution imaging of the Sun in EUV and magnetic field
DSCOVR	NOAA / NASA	2015	L1	Operational solar-wind monitor
GOES-R series	NOAA	2016–	GEO	X-ray flux (flare classes), EUV imaging, particles, magnetometer; CCOR-1 operational coronagraph on GOES-19
Parker Solar Probe	NASA	2018	Solar orbit to 6.1 million km	Origin of the solar wind
Solar Orbiter	ESA / NASA	2020	Inclined solar orbit	Close-up and polar views of the Sun
Aditya-L1	ISRO	2023	L1	India's first solar observatory
Proba-3	ESA	2024	High elliptical orbit	Formation-flying coronagraph (two spacecraft 150 m apart)

Mission	Agencies	Launch	Location	Role
PUNCH	NASA	2025	LEO (sun-synchronous)	Wide-field imaging of the corona and solar wind
SWFO-L1, IMAP	NOAA, NASA	2025	L1	Next operational monitor with coronagraph; interstellar boundary and solar-wind science
Vigil	ESA	≈ 2031	L5	Side view of the Sun–Earth line for earlier CME warning

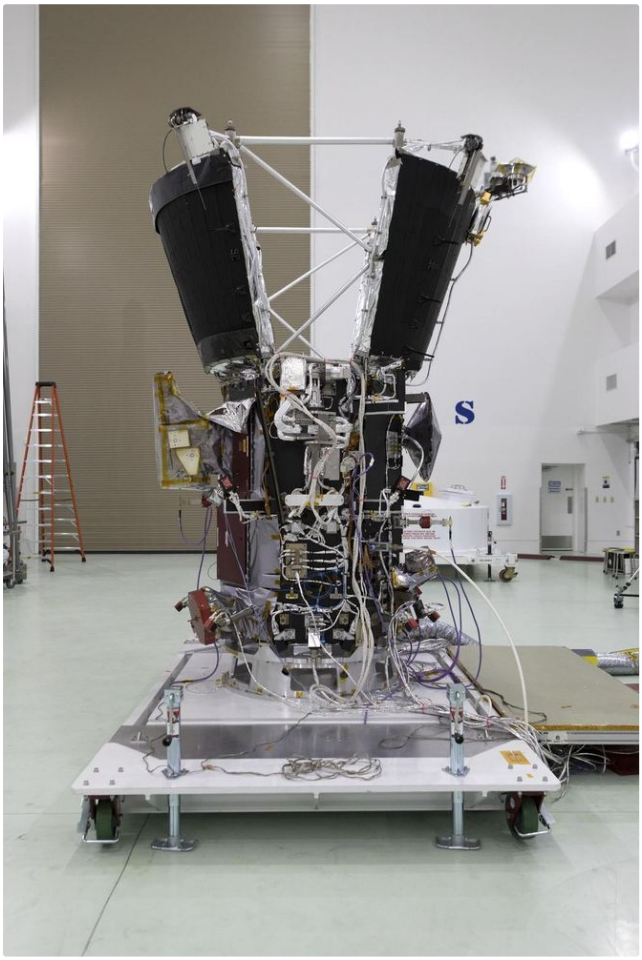
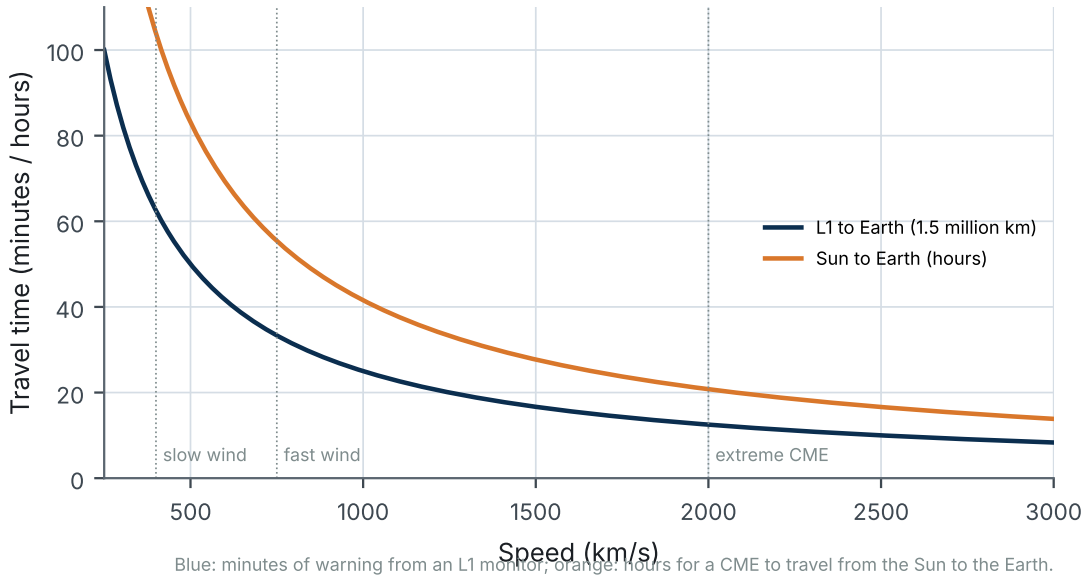


FIGURE 22.13 NASA's Parker Solar Probe during testing in 2018. Its carbon-composite heat shield allows it to fly closer to the Sun than any previous spacecraft. Image: NASA/KSC (NASA/Kim Shiflett).



FIGURE 22.14 One of the four PUNCH microsatellites during solar-array deployment tests in January 2025. Together they image the outer corona and the solar wind as a single "virtual instrument". Image: NASA/KSC (USSF 30th Space Wing/Alex Valdez).

The **L1 Lagrange point**, 1.5 million km sunward of the Earth, is the key location: a spacecraft there measures the solar wind, including the critical direction of its magnetic field, before it reaches the Earth — and an extreme CME as little as 15 minutes — this provides only short-term warning (Figure 22.15). Ground-based magnetometers, ionosondes, GNSS networks, neutron monitors and radio telescopes complete the observing system.



Blue: minutes of warning from an L1 monitor, orange: hours for a CME to travel from the Sun to the Earth.

FIGURE 22.15 Travel time from L1 to the Earth (minutes) and from the Sun to the Earth (hours) as a function of solar-wind or CME speed.

22.6.2 Forecasting

Operational space-weather centres include NOAA's **Space Weather Prediction Center** in Boulder, the UK Met Office Space Weather Operations Centre and ESA's Space Weather Service Network. Others are the regional warning centres of the International Space Environment Service. These centres issue forecasts, watches, warnings and alerts. Their tools include:

- **CME arrival forecasts**: coronagraph images are used to fit a CME's speed and direction, and a heliospheric model such as **WSA–Enlil** propagates it to the Earth. Arrival-time errors are typically ±10 hours.
- **Geomagnetic forecasts**: once a CME reaches L1, measured solar-wind conditions drive magnetospheric models (such as NOAA's Geospace model) that predict Kp, regional magnetic disturbances and GIC risk tens of minutes ahead.
- **Flare and SEP forecasts**: based mainly on the magnetic complexity of active regions; these remain probabilistic and of limited skill.
- **Ionosphere and thermosphere models** for GNSS users and satellite operators.

The greatest limitation is that the magnetic orientation of a CME, which determines its geomagnetic impact, cannot yet be measured until it reaches L1. Missions viewing the Sun from the side (L5), new methods of measuring coronal magnetic fields and machine-learning models are among the approaches to improve forecasts.

22.6.3 Space Weather Beyond the Earth

Space weather affects all planets. During the May 2024 storms, NASA's MAVEN spacecraft observed global aurorae on Mars, and its Curiosity rover measured a radiation spike on the surface (Figure 22.16). Mars lacks a global magnetic field, so it is fully exposed to SEPs — a key concern for future crews.

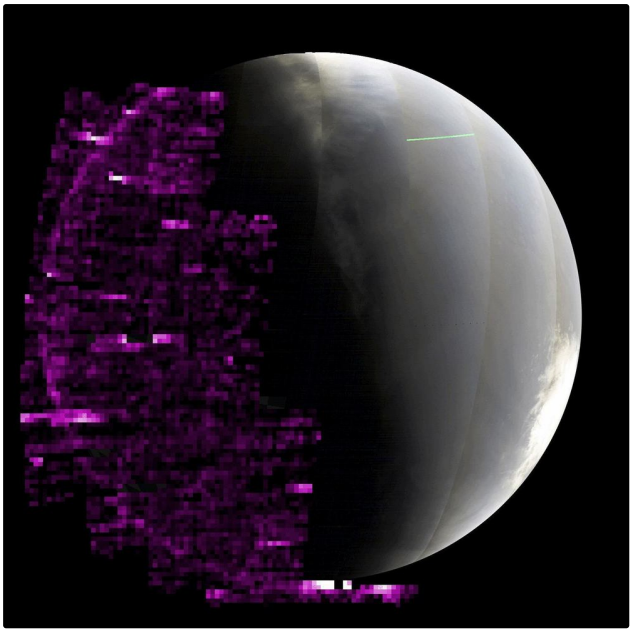


FIGURE 22.16 Ultraviolet aurorae across the night side of Mars (purple) observed by NASA's MAVEN orbiter during the solar storm of May 2024. Image: NASA/JPL (NASA/University of Colorado/LASP).

22.7 Designing for Space Weather

Engineers design spacecraft and infrastructure for specified space-weather environments:

- **Environment specifications** such as ECSS-E-ST-10-04 and NASA and ISO standards define radiation-belt fluxes, SEP fluences, charging environments and thermospheric density ranges, often including a "worst case" or 1-in-100-year event.
- **Spacecraft design**: radiation-tolerant components and error-correcting memories; shielding of sensitive parts; grounding and conductive coatings to prevent charging; derating of solar arrays for degradation; propellant margins for higher drag; autonomous safe modes.
- **Operations**: delaying launches, EVAs or critical manoeuvres during storms; placing satellites in safe configurations; increasing tracking and conjunction screening.
- **Ground infrastructure**: GIC-blocking devices and transformer designs for power grids; GNSS receivers with multi-frequency and integrity monitoring; alternative timing sources.

EXAMPLE: ESTIMATING A GEOMAGNETICALLY INDUCED CURRENT

During a severe storm, a ground magnetometer in northern Europe records magnetic variations of about 1 000 nT with a period of about 10 minutes ($\omega = 2\pi/600\ \text{s} \approx 0.0105\ \text{s}^{-1}$). For a plane wave over uniform ground of conductivity $\sigma = 10^{-3}$ S/m, the induced electric field is

$$E \approx B\sqrt{\frac{\omega}{\mu_0 \sigma}} = 10^{-6}\ \text{T} \times \sqrt{\frac{0.0105}{1.26 \times 10^{-6} \times 10^{-3}}} \approx 2.9 \times 10^{-3}\ \text{V/m} \approx 3\ \text{V/km}.$$

Over more resistive rock, or during faster variations, the field can be several times larger.

Taking E = 5 V/km along a 200 km high-voltage line, the driving voltage is 1 000 V. The loop consists of the line, two transformer windings and two substation earthings. If its total resistance is about 5 Ω, a quasi-DC current of about 200 A flows through each transformer neutral. This is enough to drive a large transformer into half-cycle saturation, causing heating, reactive-power demand and harmonics. Grid operators therefore monitor neutral currents and may reduce transformer loading during severe storms.

AI IN SPACE WEATHER

Machine learning is used to predict flares from magnetograms, estimate CME arrival times, forecast geomagnetic indices from solar-wind data, nowcast GIC and ionospheric scintillation, and model thermospheric density for satellite drag. NASA and NOAA have explored deep-learning models that predict regional geomagnetic disturbances about 30 minutes ahead from L1 data. Chapter 47 discusses these applications.

KEY TAKEAWAYS

- Space weather comprises solar flares (X-rays in 8 minutes), energetic particles (minutes to hours) and CMEs (1–3 days) that disturb the Earth's space environment.
- The Sun's magnetic activity follows an ~11-year sunspot cycle; Cycle 25 peaked at a smoothed sunspot number of about 161 in October 2024, well above forecast.
- Geomagnetic storms are measured by Kp, Ap and Dst and classified by NOAA's G scale; the May 2024 Gannon storm was the first G5 since 2003.
- Effects include satellite charging, radiation damage and drag (38 Starlink satellites lost in 2022), GNSS errors and scintillation, HF blackouts, aviation and astronaut radiation, and geomagnetically induced currents in power grids (Quebec, 1989).
- L1 monitors give 15–60 minutes of warning of a storm's severity; coronagraphs give 1–3 days of warning of CME arrival but not of its magnetic orientation.
- New observatories launched in 2024–2025 (Proba-3, PUNCH, SWFO-L1, IMAP) and a planned L5 mission (Vigil) aim to improve forecasts.

REVIEW QUESTIONS

1. Order flare X-rays, solar energetic particles and coronal mass ejections by how quickly they reach the Earth, and explain the difference.
2. A CME travels from the Sun to the Earth at an average speed of 1 500 km/s. How long does it take to arrive?
3. A solar-wind monitor at L1 is 1.5 million km upstream of the Earth. How much warning does it give for solar wind travelling at 500 km/s?
4. Explain the Kp, Ap and Dst indices and the NOAA G scale. What made the May 2024 storm notable?
5. Why did 38 Starlink satellites re-enter shortly after launch in February 2022?

6. How do geomagnetic storms induce currents in power grids, and which grids are most at risk?
7. Why can coronagraphs predict when a CME will arrive but not how strong the resulting storm will be?
8. Describe three design measures that make a satellite tolerant of space weather.

CHAPTER

23 Magnetospheric and Ionospheric Physics



Image: NASA/JSC

Every satellite in Earth orbit flies through geospace — the region dominated by the Earth's magnetic field and by electrically charged gas. This chapter explains the physics of this environment: the geomagnetic field and the magnetosphere it carves out of the solar wind; how charged particles gyrate, bounce and drift to form the Van Allen radiation belts; how magnetic reconnection couples the solar wind to the magnetosphere and drives storms, substorms and the aurora; and how solar radiation creates the ionosphere, whose layers reflect radio waves and delay navigation signals. Computed models — field lines, particle periods, ionospheric profiles and a synthetic ionogram — illustrate the key ideas, and the chapter describes the observatories, models and missions, from Explorer 1 to SMILE (launched in 2026), that have revealed geospace.

IN THIS CHAPTER

23.1 Geospace
23.2 The Geomagnetic Field
23.3 The Magnetosphere
23.4 Charged Particles in Magnetic Fields
23.5 Coupling the Solar Wind to the Magnetosphere
23.6 The Aurora
23.7 The Upper Atmosphere
23.8 The Ionosphere
23.9 Measuring and Modelling Geospace

23.1 Geospace

Above about 60 km, the Earth's atmosphere becomes partly ionised by solar radiation. Above about 1 000 km, the environment is dominated by **plasma** — a gas of free electrons and ions. This plasma is controlled by electric and magnetic fields rather than by collisions. This region, from

the ionosphere out to the boundary with the solar wind, is called **geospace**. It contains the radiation belts that damage satellites (Chapter 2) and the ionosphere that affects every radio signal to and from space (Chapters 7 and 18). It also contains the thermosphere, whose drag determines satellite lifetimes (Chapters 11 and 21). Understanding its physics is the basis of space weather forecasting (Chapter 22).

23.2 The Geomagnetic Field

23.2.1 A Tilted Dipole

The Earth's magnetic field is generated by a dynamo in the liquid iron outer core. At the surface it is approximately that of a **magnetic dipole** tilted by about 10° from the rotation axis. Its field strength is about 30 μT at the equator and 60 μT near the poles. In a dipole, the field strength falls with distance as r^{-3}, and field lines follow

$$r = L\cos^2\lambda$$

where λ is magnetic latitude and L is the distance, in Earth radii (R_E), at which the field line crosses the magnetic equator (Figure 23.1). The **L-shell** parameter is the most convenient way to label regions of the magnetosphere. The inner radiation belt lies at L ≈ 1.2–2, GNSS satellites at L ≈ 4–4.5, and geostationary satellites at L ≈ 6.6.

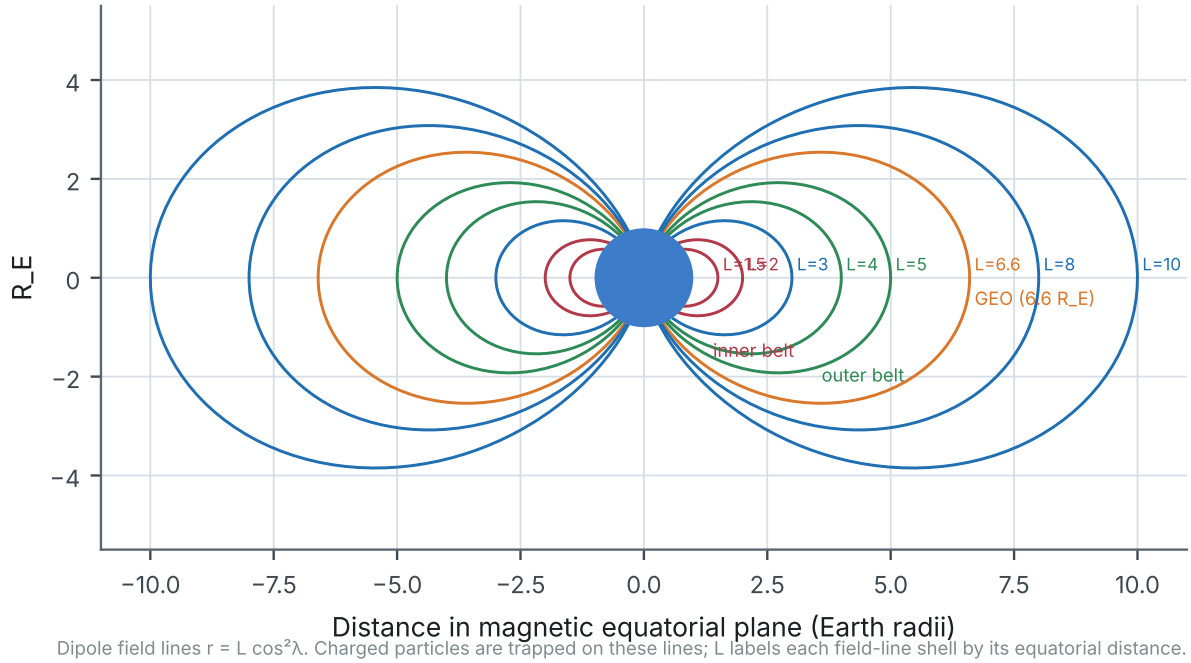


FIGURE 23.1 Dipole magnetic field lines labelled by L-shell. Trapped particles spiral along these lines; geostationary satellites are on the L = 6.6 shell.

23.2.2 A Changing Field

The field is not static. The north magnetic pole has drifted from northern Canada towards Siberia at up to 50 km per year. The dipole moment has weakened by about 9 % since the mid-19th century. Over geological time the field has reversed many times — most recently about 780 000 years ago. A region of weak field over South America and the South Atlantic, the **South Atlantic Anomaly**, allows the inner radiation belt to dip to altitudes of 200–800 km. LEO satellites —

including the ISS and the Hubble Space Telescope — pass through it several times a day. They experience most of their radiation dose and single-event upsets there. The field is monitored by ground observatories and by ESA's three **Swarm** satellites (2013). It is modelled by the International Geomagnetic Reference Field (IGRF) and the World Magnetic Model. These models are updated every five years and used in every smartphone compass and aircraft heading system.

23.3 The Magnetosphere

23.3.1 Structure

The supersonic solar wind cannot penetrate the Earth's magnetic field. Instead, the field carves out a cavity — the **magnetosphere** — shaped like a comet, compressed on the dayside and stretched into a long tail on the nightside (Figure 23.2):

- The **bow shock**, about 13 R_E upstream, where the solar wind slows abruptly from supersonic to subsonic speed;
- the **magnetosheath** of shocked, turbulent solar-wind plasma;
- the **magnetopause**, the boundary where the pressure of the solar wind balances the magnetic pressure of the Earth's field — typically about 10 R_E from the Earth on the sunward side;
- the polar **cusps**, funnel-shaped regions through which solar-wind plasma can reach the upper atmosphere;
- the **magnetotail**, extending more than 200 R_E downstream, consisting of two **lobes** of nearly empty, open field lines separated by the hot **plasma sheet**;
- the inner magnetosphere: the cold, dense **plasmasphere** (an extension of the ionosphere to about 4–5 R_E), the **radiation belts** and the **ring current** of energetic ions that flows around the Earth during storms.

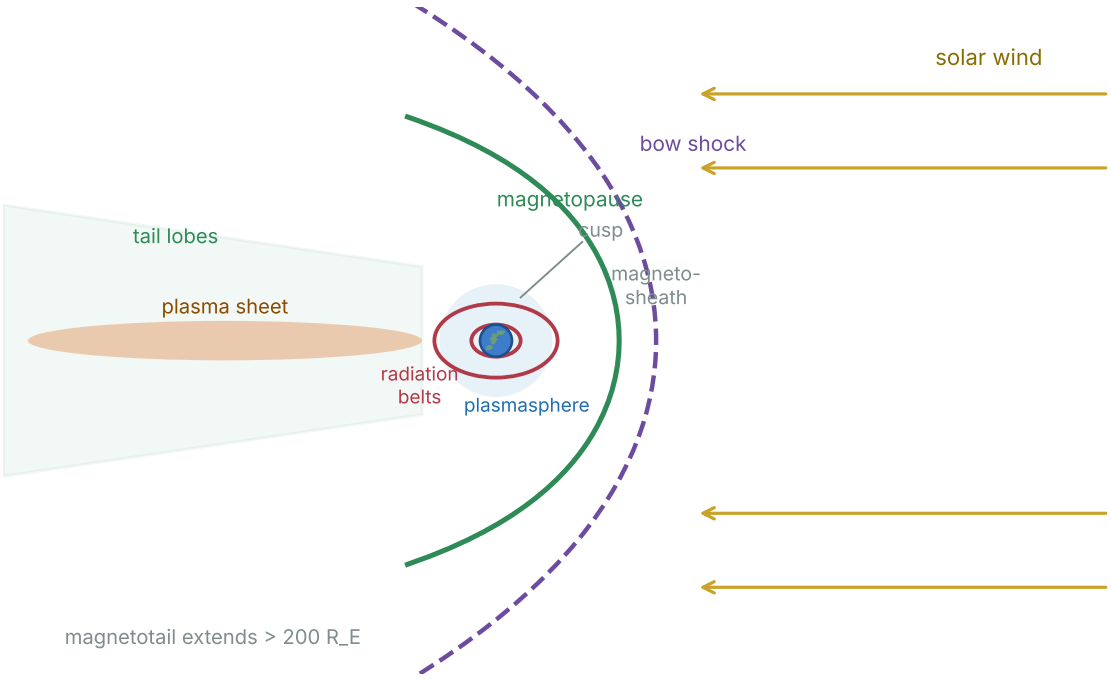


FIGURE 23.2 Structure of the Earth's magnetosphere (schematic, not to scale). The Sun is to the right.

23.3.2 Where Is the Magnetopause?

The distance to the magnetopause follows from pressure balance. The solar wind's dynamic pressure $P = \rho v^2$ must equal the magnetic pressure of the compressed dipole field $B^2/2\mu_0$; allowing for a factor of about 2 compression by the boundary currents, the standoff distance is

$$r_{mp} \approx \left(\frac{2B_0^2}{\mu_0 P} \right)^{1/6} R_E$$

with $B_0 \approx 31$ μT the equatorial surface field. For a typical pressure of 2 nPa this gives $r_{mp} \approx 9.6\,R_E$; because of the sixth root, even large pressure increases move it only modestly (Figure 23.3). During extreme storms, high pressure combined with erosion by a southward interplanetary field pushes the magnetopause inside geostationary orbit, exposing GEO satellites directly to the solar wind and the magnetosheath.

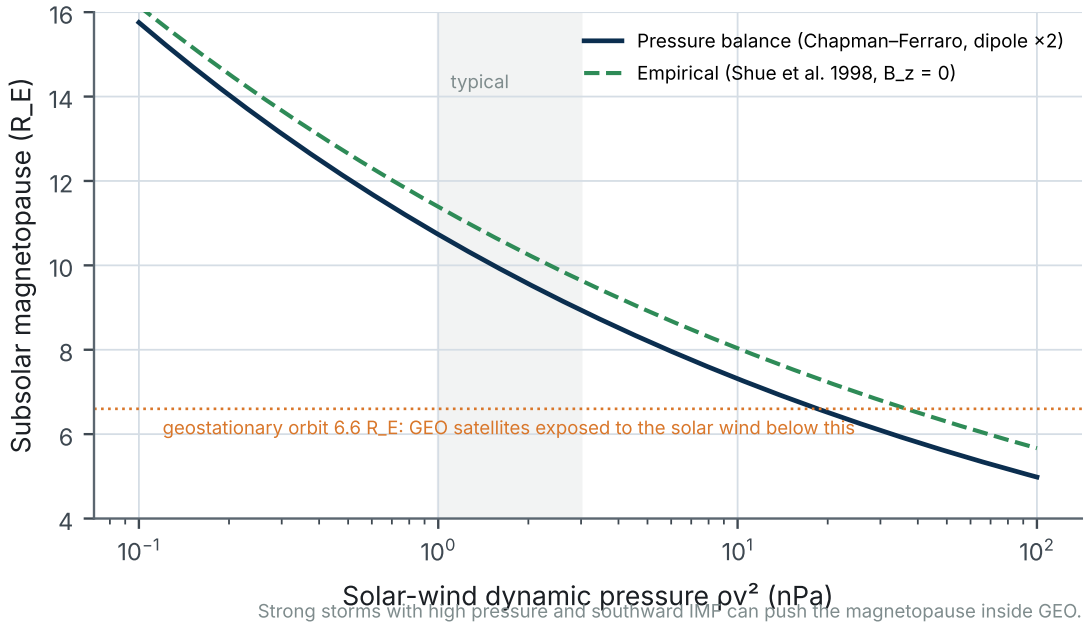


FIGURE 23.3 Subsolar magnetopause distance versus solar-wind dynamic pressure, from pressure balance and from an empirical model.

23.4 Charged Particles in Magnetic Fields

23.4.1 Three Periodic Motions

A charged particle in a magnetic field performs three superimposed periodic motions (Figure 23.4):

1. **Gyration** around the field line, with the **gyrofrequency** $f_g = qB/(2\pi\gamma m)$ — tens of kilohertz for electrons and a few hertz for protons in the inner magnetosphere.
2. **Bounce** along the field line between **mirror points**. As a particle moves towards stronger field near the poles, its velocity component perpendicular to the field grows at the expense of the parallel component (conserving its magnetic moment) until it reverses direction. Particles whose **pitch angle** is too small mirror below about 100 km altitude, where they are lost by collisions with the atmosphere — they are said to be in the **loss cone**.

3. **Drift** around the Earth, caused by the gradient and curvature of the field: electrons drift eastward and ions westward, forming the ring current.

The three periods differ by orders of magnitude (Figure 23.5): gyration in microseconds to fractions of a second, bounce in fractions of a second to seconds, drift in minutes to hours. Each motion is associated with an **adiabatic invariant** that is conserved when the fields change slowly compared with that period. Violating the invariants — through waves, sudden compressions or electric fields — accelerates, transports and removes particles.

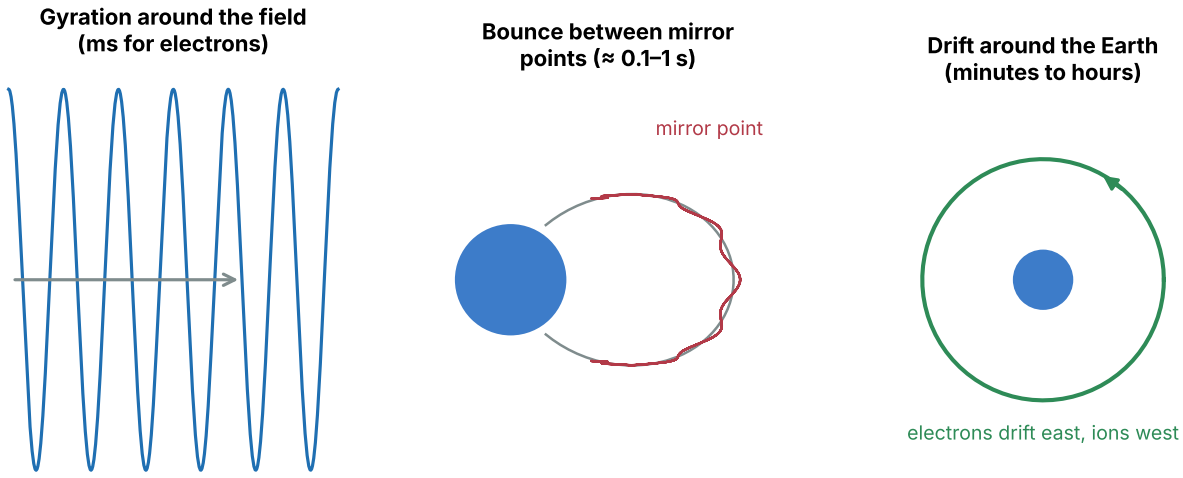
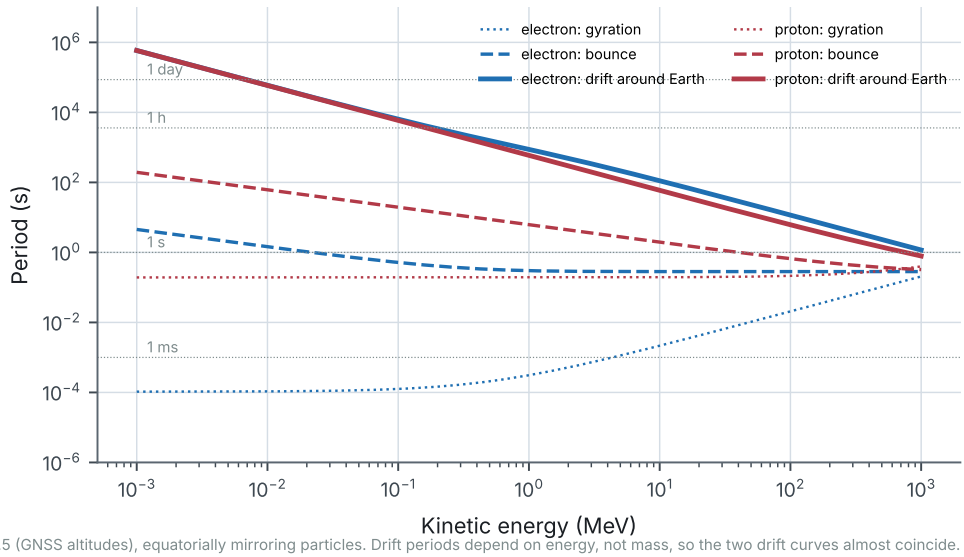


FIGURE 23.4 The three periodic motions of trapped particles: gyration, bounce between mirror points and drift around the Earth.



Dipole field at L = 4.5 (GNSS altitudes), equatorially mirroring particles. Drift periods depend on energy, not mass, so the two drift curves almost coincide.

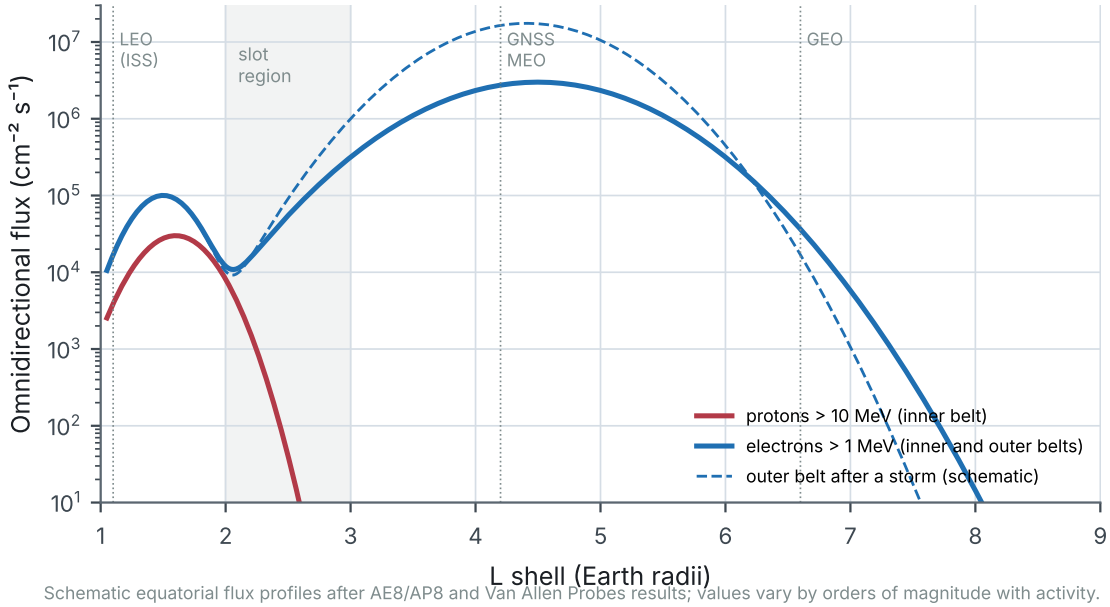
FIGURE 23.5 Gyration, bounce and drift periods of electrons and protons as a function of energy at L = 4.5.

23.4.2 The Van Allen Radiation Belts

The first US satellite, **Explorer 1** (1958), carried a Geiger counter designed by James Van Allen of the University of Iowa. It recorded puzzling drop-outs — later understood as saturation by intense radiation — revealing that the Earth is surrounded by belts of trapped energetic particles. The **radiation belts** (Figure 23.6) consist of:

- An **inner belt** at L ≈ 1.2–2, dominated by protons of tens to hundreds of MeV, created mainly by the decay of neutrons produced by cosmic rays hitting the atmosphere (cosmic-ray albedo neutron decay). It is very stable.
- An **outer belt** at L ≈ 3–7, dominated by electrons of hundreds of keV to several MeV, which varies by orders of magnitude within hours during storms, as wave–particle interactions accelerate and scatter electrons.
- A **slot region** at L ≈ 2–3, kept relatively empty by scattering of electrons into the loss cone by plasmaspheric waves.

NASA's twin **Van Allen Probes** (2012–2019) transformed understanding of the belts. Within days of launch they discovered a transient **third belt** (Figure 23.7). They also showed that electrons are accelerated locally by "chorus" waves inside the belts, and found a sharp barrier at about 11 000 km that ultra-relativistic electrons cannot cross. Engineering models of the trapped radiation environment, such as AE8/AP8 and their successors AE9/AP9/SPM, are used to design spacecraft (Chapter 2).



Schematic equatorial flux profiles after AE8/AP8 and Van Allen Probes results; values vary by orders of magnitude with activity.

FIGURE 23.6 Schematic radial profiles of trapped protons and electrons in the radiation belts. The outer electron belt varies greatly with geomagnetic activity.

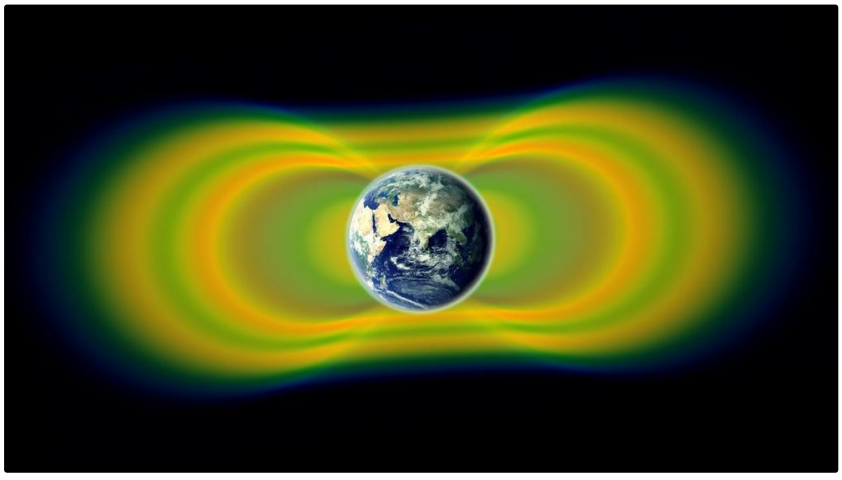


FIGURE 23.7 The Van Allen Probes discovered in September 2012 that a third, transient radiation belt (in the middle) can form between the inner and outer belts. Visualisation by NASA. Image: NASA/GSFC.

The belts can also be created artificially. The US **Starfish Prime** nuclear test in July 1962, at 400 km altitude, injected high-energy electrons that formed an artificial belt lasting years. It damaged or destroyed about a third of the satellites then in orbit — one reason why nuclear explosions in space are now banned (Chapter 19).

23.5 Coupling the Solar Wind to the Magnetosphere

23.5.1 Magnetic Reconnection

How does solar-wind energy enter a magnetosphere that the solar wind cannot penetrate? The answer, proposed by James Dungey in 1961, is **magnetic reconnection**. When the interplanetary magnetic field points southward — opposite to the Earth's field at the dayside magnetopause — field lines from the two regions can break and reconnect. Terrestrial field lines then become connected to the solar wind (Figure 23.8). These "open" field lines are swept back over the poles by the solar wind and pile up in the tail lobes. There they reconnect again in the plasma sheet, and the closed field lines return to the dayside. This **Dungey cycle** drives a large-scale circulation of plasma — two convection cells in the polar ionosphere — and transfers energy, mass and momentum from the solar wind into the magnetosphere. When the interplanetary field is northward, reconnection occurs at high latitudes instead and the magnetosphere is relatively quiet.

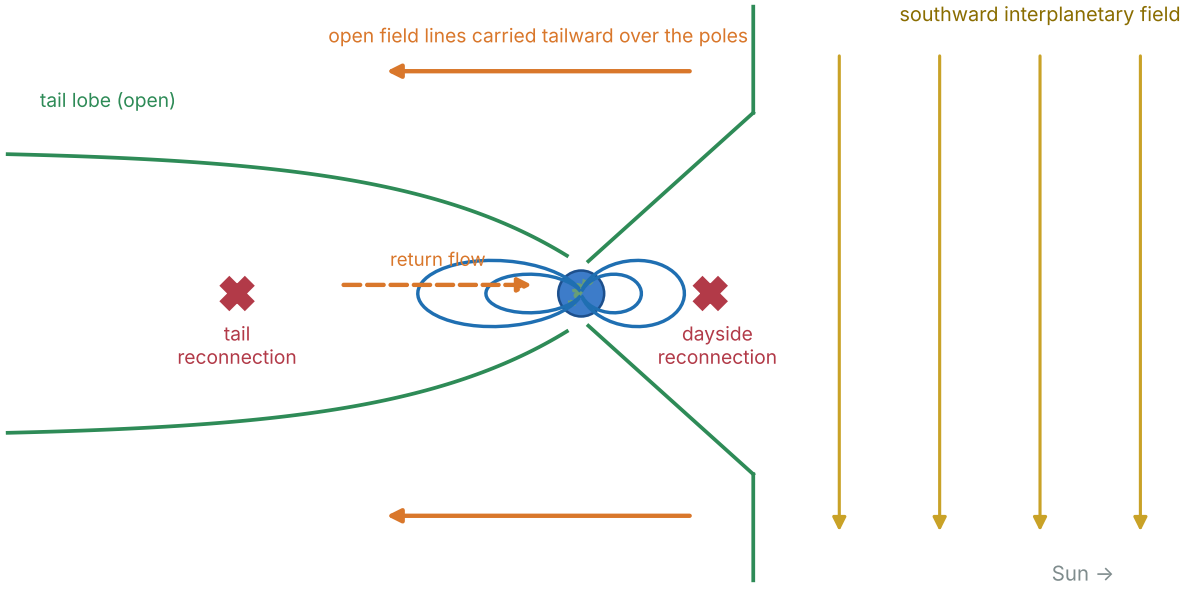


FIGURE 23.8 The Dungey cycle. Reconnection with a southward interplanetary magnetic field opens dayside field lines, which are carried over the poles into the tail, where they reconnect and return.

Reconnection converts magnetic energy into particle energy in tiny regions, a few kilometres across for electrons, where the plasma decouples from the magnetic field. NASA's four **Magnetospheric Multiscale (MMS)** spacecraft were launched in 2015 and fly in a tetrahedron only a few kilometres across. They were the first to fly through these electron diffusion regions and measure the physics of reconnection directly (Figure 23.9). The same physics also operates in solar flares, fusion reactors and astrophysical plasmas.

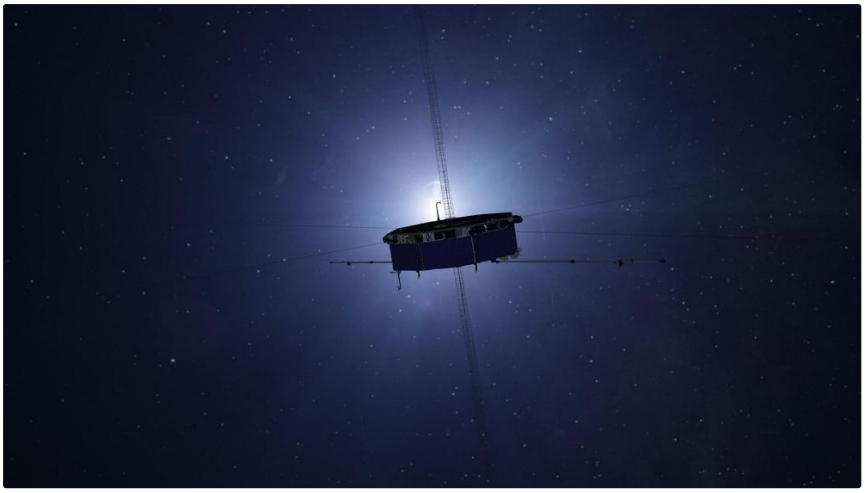


FIGURE 23.9 Artist's impression of NASA's four Magnetospheric Multiscale spacecraft flying in formation through a region of magnetic reconnection. Image: NASA/GSFC.

23.5.2 Substorms and Storms

Energy stored in the stretched tail is released intermittently in **substorms**. Over about an hour, the tail field becomes increasingly stretched (growth phase). Then reconnection in the near-Earth tail suddenly releases energy (expansion phase), and the tail field snaps back towards a dipole. Particles are injected into the inner magnetosphere, and the aurora brightens and surges poleward and westward. Substorms occur several times a day even in moderately disturbed conditions; NASA's five **THEMIS** satellites (2007) were designed to determine where and how they start.

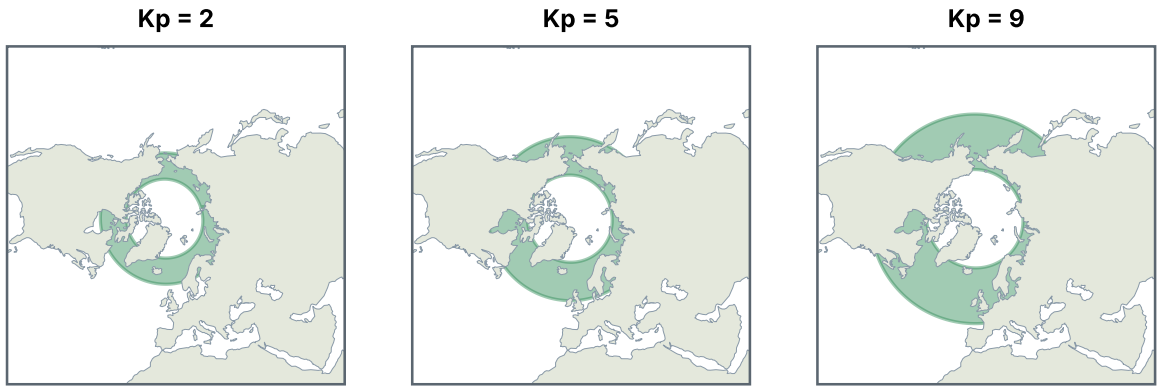
A **geomagnetic storm** is a longer, larger disturbance lasting one to several days, driven by a sustained period of strong southward interplanetary field — usually from a CME or a corotating interaction region (Chapter 22). During a storm, ions injected from the tail build up the **ring current**, whose magnetic field depresses the horizontal field at low latitudes (measured by the Dst index). The radiation belts are also reconfigured, and large currents flow along field lines into the polar ionosphere (**field-aligned** or Birkeland currents). These currents close through the auroral electrojets.

23.6 The Aurora

The aurora is produced when electrons with energies of about 1–10 keV, accelerated in the magnetosphere above the ionosphere, plunge along field lines into the upper atmosphere. There they excite oxygen atoms and nitrogen molecules, which emit light as they return to lower energy states:

- **Green** light at 557.7 nm from atomic oxygen, at about 100–250 km — the most common auroral colour;
- **Red** light at 630.0 nm from atomic oxygen, at higher altitudes (above about 200 km), where collisions are rare enough for this slow transition to occur;
- **Blue and violet** light from ionised molecular nitrogen (391.4 and 427.8 nm) at lower altitudes, often seen at the bottom edge of active displays.

Aurorae occur in two **auroral ovals** centred on the geomagnetic poles, typically between about 65° and 75° magnetic latitude. During storms the ovals expand towards the equator (Figure 23.10): in the May 2024 storm, aurorae were seen at magnetic latitudes below 30°. Astronauts on the ISS, whose orbit reaches 51.6° latitude, regularly fly through the auroral zones (Figure 23.11). Space-based ultraviolet imagers have photographed the entire oval from above since the 1980s.



Schematic auroral oval (geographic coordinates, approximate). The oval expands towards the equator as activity increases.

FIGURE 23.10 Schematic auroral ovals for quiet (Kp 2), moderate (Kp 5) and extreme (Kp 9) geomagnetic activity.



FIGURE 23.11 The aurora australis above the Indian Ocean, photographed from the International Space Station on 12 June 2025 between Australia and Antarctica. Image: NASA/JSC.

23.7 The Upper Atmosphere

Above about 90 km, the neutral atmosphere changes character (Figure 23.12). In the **thermosphere**, absorption of solar extreme-ultraviolet radiation heats the gas to 600–2 000 K, depending on solar activity. Molecules are dissociated, so **atomic oxygen** becomes the dominant species between about 200 and 600 km. This atomic oxygen causes material erosion on spacecraft (Chapter 2). Above about 600–1 000 km, in the **exosphere**, helium and then hydrogen dominate and atoms move on ballistic paths, sometimes escaping to space. Because the thermosphere

expands when heated, its density at a given altitude varies by a factor of ten or more over the solar cycle. Its density also increases several-fold during storms. This is the main reason for uncertainty in predicting satellite orbits (Chapters 21 and 22). NASA's TIMED (2001), GOLD (2018) and ICON (2019–2022) missions have studied how the thermosphere responds both to the Sun from above and to atmospheric waves from below.

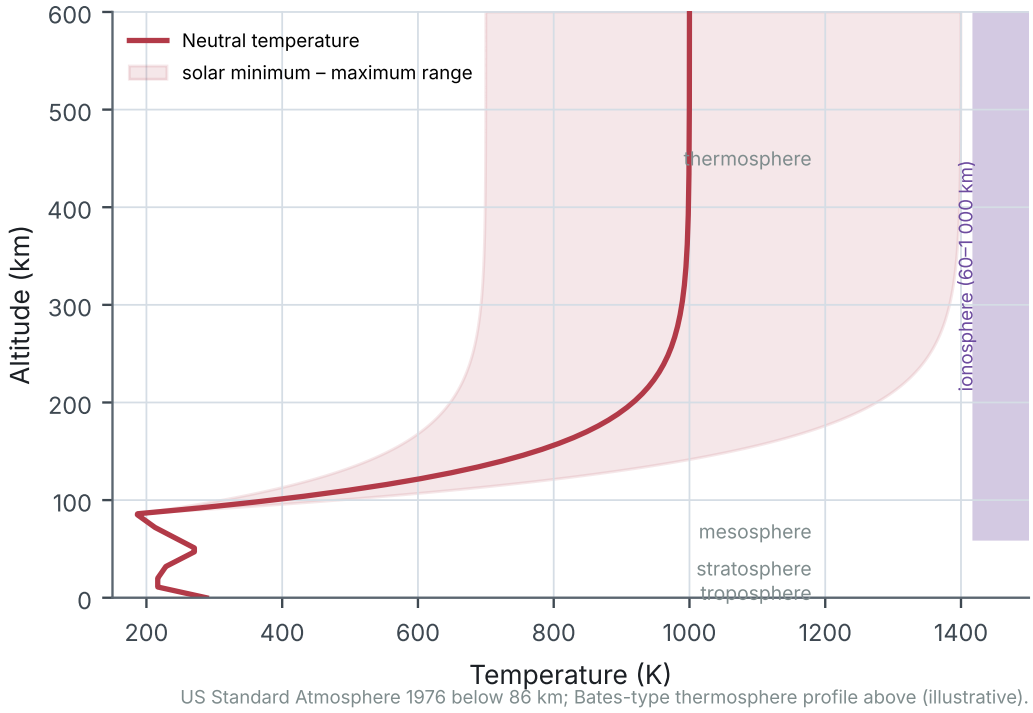


FIGURE 23.12 Temperature of the neutral atmosphere with altitude: US Standard Atmosphere below 86 km and a thermospheric profile above, with the range between solar minimum and maximum.

23.8 The Ionosphere

23.8.1 Formation and Layers

Solar extreme-ultraviolet and X-ray radiation ionises the upper atmosphere, creating the **ionosphere** between about 60 and 1 000 km. Its electron density at each altitude results from a balance between production, which is greatest where the radiation is absorbed (a Chapman layer), and loss by recombination and transport. The ionosphere has distinct layers (Figure 23.13):

- the **D region** (60–90 km), present only in daytime, weakly ionised but dense enough in neutral gas to absorb HF radio waves;
- the **E region** (90–150 km), with a peak near 110 km and occasional thin, intense layers of metallic ions ("sporadic E");
- the **F1 layer** (150–220 km), present in daytime;
- the **F2 layer** (220–500 km), the densest, with peak densities of about 10^{11}–10^{12} electrons per m^3, which persists at night because recombination is slow at high altitudes.

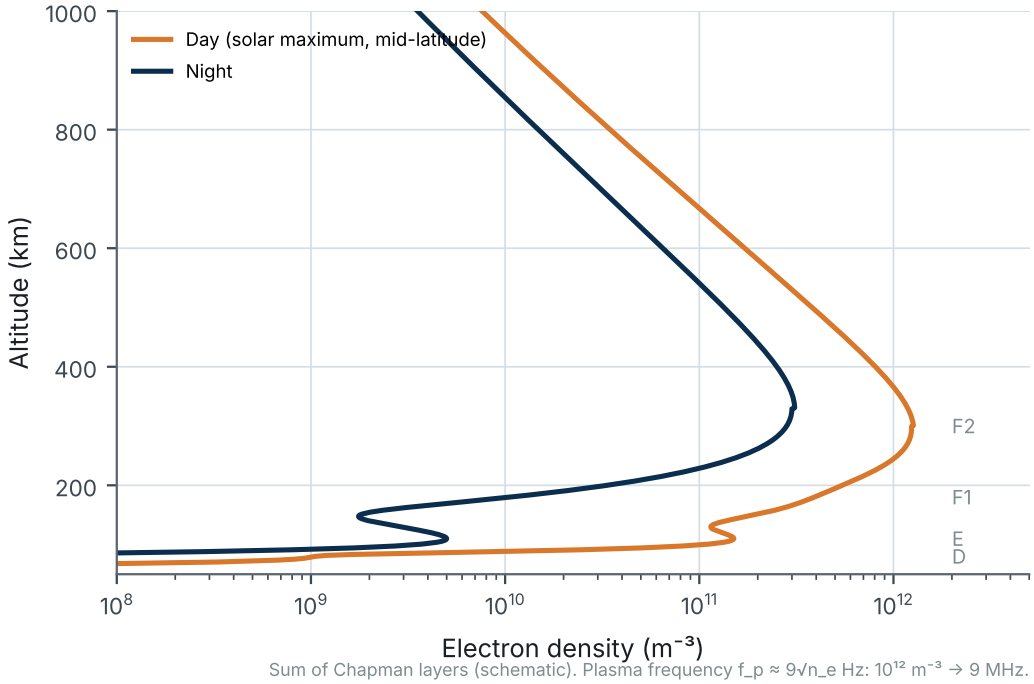


FIGURE 23.13 Electron density profiles of the ionosphere by day and by night (schematic Chapman-layer model).

23.8.2 Variations and Irregularities

The ionosphere varies with time of day, season, solar cycle (peak densities are several times higher at solar maximum) and geomagnetic activity. Near the magnetic equator, the eastward electric field of the daytime ionosphere lifts plasma upward ("fountain effect"). The plasma then slides down the field lines to form two crests of high density about 15° north and south of the magnetic equator. This is the **equatorial ionization anomaly**, where total electron content reaches its global maximum (Figure 23.14). After sunset, the bottom of the equatorial F region can become unstable, forming **plasma bubbles** — depletions that rise through the ionosphere and fill with irregularities of all scales, causing strong **scintillation** of satellite signals. At high latitudes, polar-cap patches, auroral precipitation and travelling ionospheric disturbances create other irregularities.

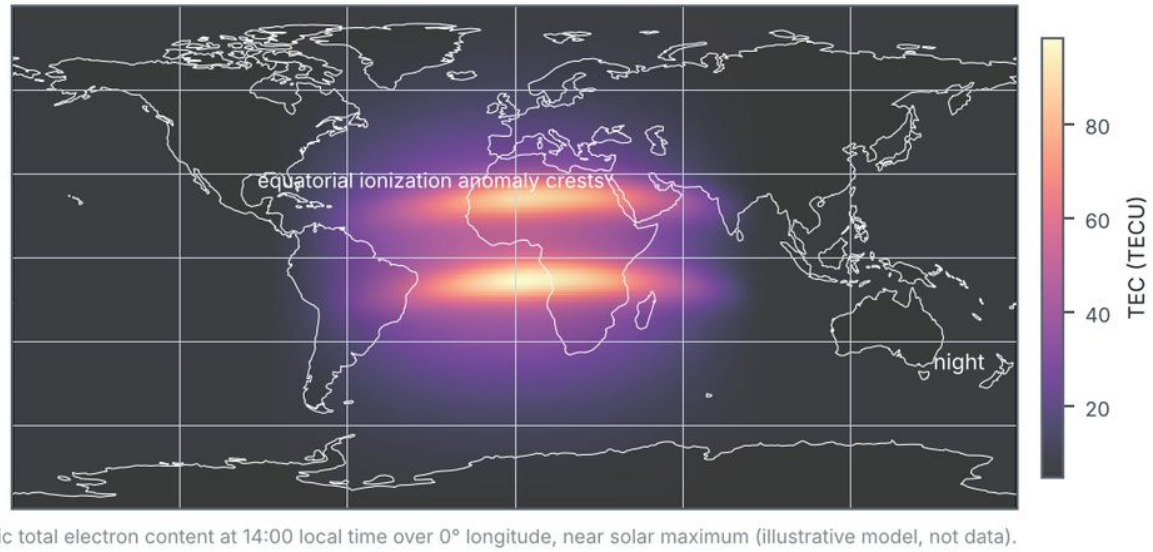


FIGURE 23.14 Schematic global total electron content in the afternoon near solar maximum, showing the crests of the equatorial ionization anomaly.

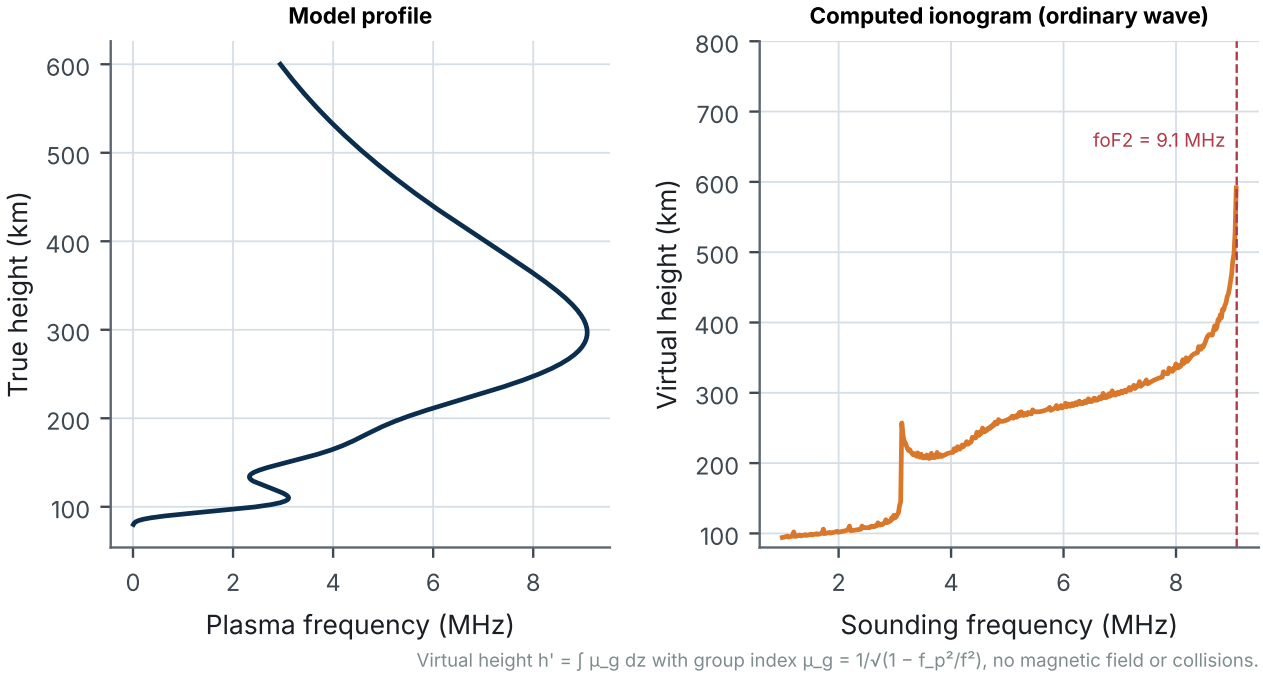
23.8.3 Radio Propagation

The ionosphere's effect on radio waves depends on the **plasma frequency**

$$f_p \approx 8.98\sqrt{n_e}\ \text{Hz}$$

where n_e is the electron density in m^{-3}. Radio waves below the local plasma frequency cannot propagate and are reflected; waves above it pass through but are delayed and refracted.

- **HF skywave propagation**: waves at 3–30 MHz are reflected by the ionosphere, allowing over-the-horizon communication and broadcasting worldwide. The highest frequency reflected at vertical incidence is the **critical frequency** foF2 (5–15 MHz); at oblique incidence the **maximum usable frequency** is higher by roughly the secant of the angle of incidence.
- **Ionosondes** sweep a radio frequency upward and measure the delay of the echo, producing an **ionogram** of virtual height versus frequency (Figure 23.15), from which the electron-density profile is reconstructed. Networks of ionosondes, and over-the-horizon radars, rely on the same physics.
- **Satellite signals** at frequencies far above the plasma frequency pass through, but they are delayed. The delay is proportional to the total electron content divided by the frequency squared (40.3 TEC/f^2 metres; Chapter 18). Their polarisation is also rotated by the magnetic field (**Faraday rotation**), which affects low-frequency radars and radio astronomy.
- **Scintillation**, measured by the S4 index (amplitude) and σϕ (phase), disrupts GNSS and satellite communications at L-band and below, mainly near the equator after sunset and at high latitudes.



Virtual height h' = ∫ μ_g dz with group index μ_g = 1/√(1 − f_p²/f²), no magnetic field or collisions.

FIGURE 23.15 A model electron-density profile (left) and the ionogram computed from it (right). Virtual heights rise steeply near the critical frequencies of the E and F2 layers, where the signal is slowed most.

23.9 Measuring and Modelling Geospace

23.9.1 Observations

Geospace is studied with a combination of ground and space instruments:

- **Ground**: magnetometer networks (which provide the Kp, Dst and AE indices), ionosondes, and coherent-scatter radars (SuperDARN, which maps ionospheric convection over both polar regions). There are also **incoherent-scatter radars** (such as those at Millstone Hill, Jicamarca and the European EISCAT system, being replaced by the volumetric EISCAT_3D). Other instruments are GNSS receiver networks that map total electron content globally, all-sky cameras and riometers.
- **Space**: missions from Explorer 1 onward (Figure 23.16). They include ISEE, Dynamics Explorer, Geotail, Polar and Wind, ESA's four-spacecraft **Cluster** (2000–2024), THEMIS, the Van Allen Probes, Swarm, MMS, Japan's Arase (2016), GOLD and ICON. Radio occultation also provides ionospheric profiles (Chapter 16). NASA's **TRACERS** twin satellites (launched July 2025) study reconnection in the cusp. The ESA–Chinese Academy of Sciences mission **SMILE** was launched on 19 May 2026. It carries a soft X-ray imager that, for the first time, images the magnetopause and cusps from outside, together with an ultraviolet auroral imager.

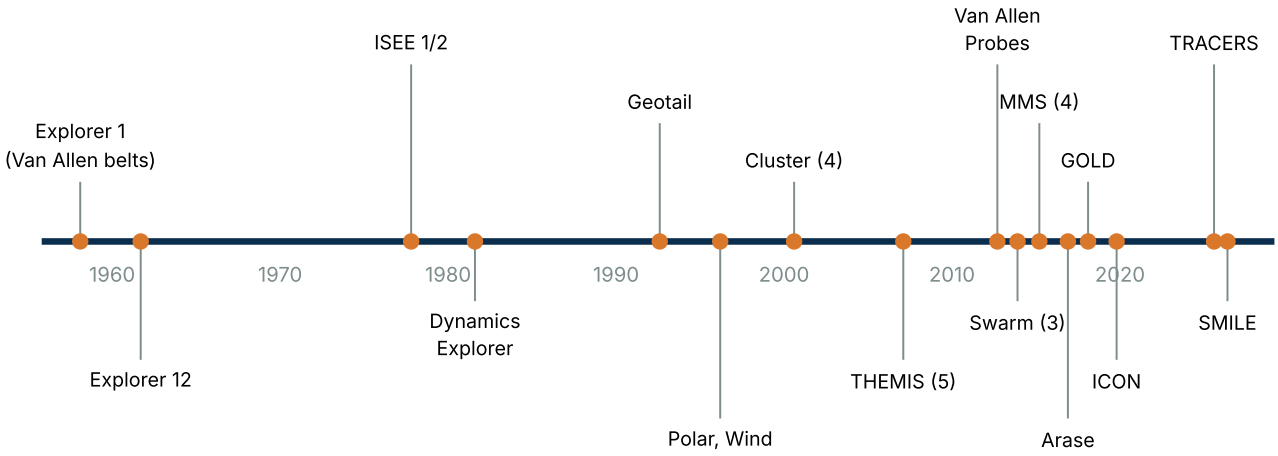


FIGURE 23.16 Selected magnetospheric and ionospheric missions.

23.9.2 Models

Engineers and forecasters use a family of standard models:

TABLE 23.1 Standard models of the geospace environment.

Model	Describes	Typical use
IGRF, World Magnetic Model	Main geomagnetic field	Navigation, attitude determination, L-shell computation
Tsyganenko and other magnetospheric field models	External field from currents	Particle tracing, mapping field lines during storms
AE9/AP9/SPM (and legacy AE8/AP8)	Trapped radiation	Radiation-dose and single-event analysis in spacecraft design

Model	Describes	Typical use
IRI (International Reference Ionosphere)	Electron density, temperatures	Radio propagation, GNSS corrections, research
Klobuchar, NeQuick	Ionospheric delay	Broadcast corrections for GPS and Galileo single-frequency users
NRLMSIS 2.0, JB2008, DTM	Thermospheric density and composition	Orbit determination and lifetime prediction
Space Weather Modeling Framework, Geospace model	Coupled Sun-to-Earth physics	Operational forecasting

23.9.3 Other Planets

Magnetospheres are common in the Solar System (Table 23.2). Jupiter's is the largest structure in the Solar System after the heliosphere — its magnetopause lies 50–100 Jovian radii from the planet. Its intense radiation belts, fed by sulphur and oxygen from Io's volcanoes, constrain the design of every Jupiter mission (Juno, JUICE, Europa Clipper). Mercury's small magnetosphere is so compressed that the solar wind sometimes reaches its surface. Mars and Venus have no global fields, so the solar wind interacts directly with their upper atmospheres and strips them away, as measured by MAVEN.

TABLE 23.2 Magnetospheres of the planets (approximate).

Planet	Global field (relative to Earth's dipole moment)	Typical magnetopause distance	Notes
Mercury	≈ 0.0006	≈ 1.5 planetary radii	Studied by MESSENGER; BepiColombo arrives in 2026
Venus	None (induced magnetosphere)	—	Ionosphere interacts directly with the solar wind
Earth	1	≈ 10 R_E	Radiation belts, aurora
Mars	None (crustal fields)	—	Atmospheric escape measured by MAVEN
Jupiter	≈ 20 000	50–100 R_J	Intense radiation belts; Io plasma torus
Saturn	≈ 580	≈ 20 R_S	Enceladus supplies water-group ions
Uranus, Neptune	≈ 50, ≈ 25	≈ 20 planetary radii	Strongly tilted, offset fields

EXAMPLE: HF COMMUNICATION ACROSS THE ATLANTIC

An airline crew over the North Atlantic must contact a ground station 3 000 km away by HF radio. At night the F2-layer peak is at about 300 km with foF2 = 5 MHz.

1. For a single-hop path of 3 000 km, the ray reflects at the midpoint, 1 500 km from each end. Allowing for the Earth's curvature, the elevation angle at the ground is only a few degrees and the angle of incidence on the layer is about 75°, so the secant factor is about 3.5–4.
2. The maximum usable frequency is therefore about 5 × 3.7 ≈ 18 MHz. Operators choose a working frequency about 15 % below this, around 15 MHz.
3. After sunrise, foF2 rises to about 10 MHz and the MUF to more than 30 MHz, but the D region also forms and absorbs lower frequencies, so the crew must move to a higher band.
4. During a polar-cap absorption event after a solar proton event (Chapter 22), D-region absorption at high latitudes can block HF entirely for days; aircraft then use satellite communications or reroute to lower latitudes.

AI IN MAGNETOSPHERIC AND IONOSPHERIC PHYSICS

Neural networks trained on decades of satellite and ground data now predict geomagnetic indices, map radiation-belt electron fluxes, forecast ionospheric total electron content and scintillation, and reconstruct three-dimensional ionospheric structure from sparse GNSS observations. Physics-informed machine learning combines these data-driven models with the conservation laws of plasma physics (Chapters 44 and 47).

KEY TAKEAWAYS

- The Earth's field is approximately a tilted dipole; field lines are labelled by L-shell. The South Atlantic Anomaly brings the inner belt down to LEO altitudes.
- The magnetosphere is bounded by the magnetopause (about 10 R_E sunward) and bow shock and has a long tail; the standoff distance follows from pressure balance and scales as P^(−1/6).
- Trapped particles gyrate, bounce and drift; the radiation belts comprise a stable inner proton belt and a highly variable outer electron belt separated by a slot.
- Magnetic reconnection with a southward interplanetary field drives the Dungey cycle, substorms and storms; MMS measured reconnection directly.
- The aurora is produced by 1–10 keV electrons exciting oxygen (green, red) and nitrogen (blue, violet) in the auroral ovals, which expand during storms.
- The ionosphere's D, E, F1 and F2 layers reflect HF radio below the plasma frequency (f_p ≈ 9√n_e Hz) and delay and scintillate satellite signals.
- Geospace is monitored by ground networks and missions from Explorer 1 to SMILE (2026), and described by standard models used throughout space engineering.

REVIEW QUESTIONS

1. The Earth's equatorial surface field is about 30 μT. Estimate the field at the equator of the L = 6.6 shell (geostationary distance) using the dipole law.
2. The magnetopause stand-off distance scales as the solar-wind dynamic pressure to the power −1/6. If the pressure rises eightfold from its typical value, how far does a 10 R_E magnetopause move inwards?
3. Describe the three periodic motions of a charged particle trapped in the Earth's magnetic field.
4. Why is the outer radiation belt much more variable than the inner belt? What is the slot region?

5. What is the South Atlantic Anomaly, and why does it matter for satellites and astronauts in low orbit?
6. Explain how magnetic reconnection with a southward interplanetary magnetic field drives geomagnetic activity.
7. Why are aurorae mostly green and red? At what altitudes is each colour produced?
8. Calculate the plasma frequency for an electron density of $10^{12}\ \mathrm{m}^{-3}$. What happens to HF radio waves below and above this frequency?